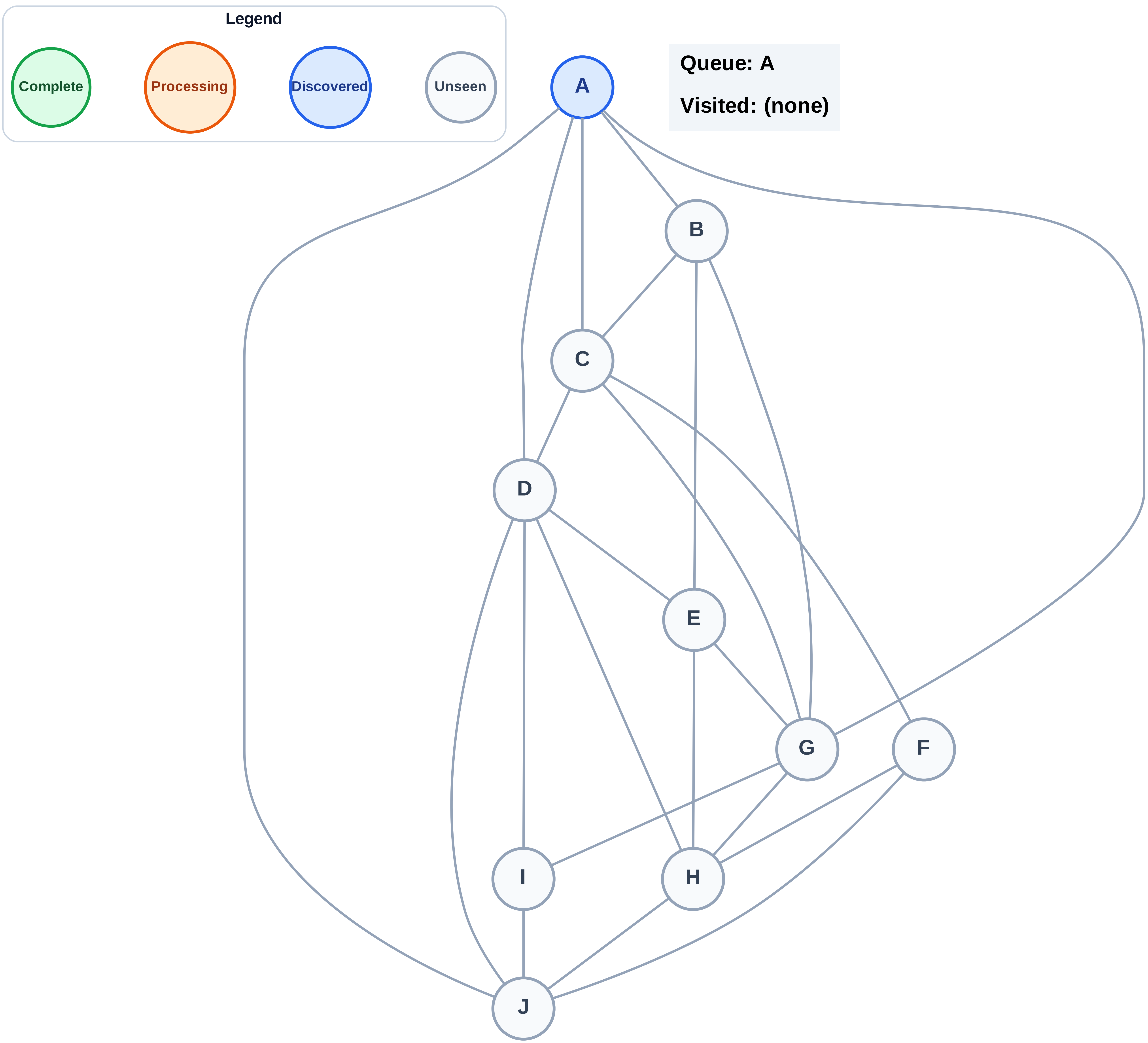
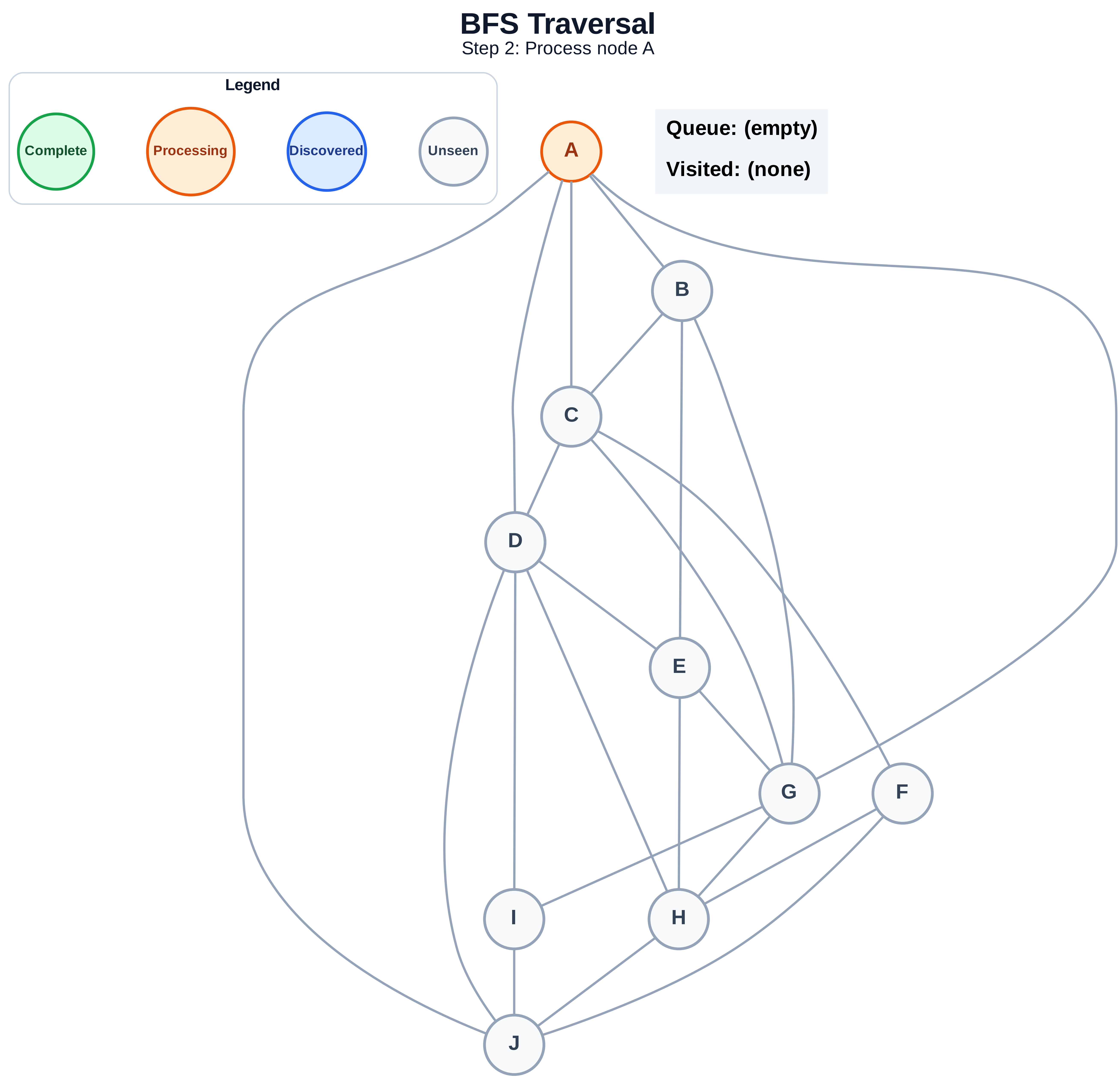
BFS Traversal

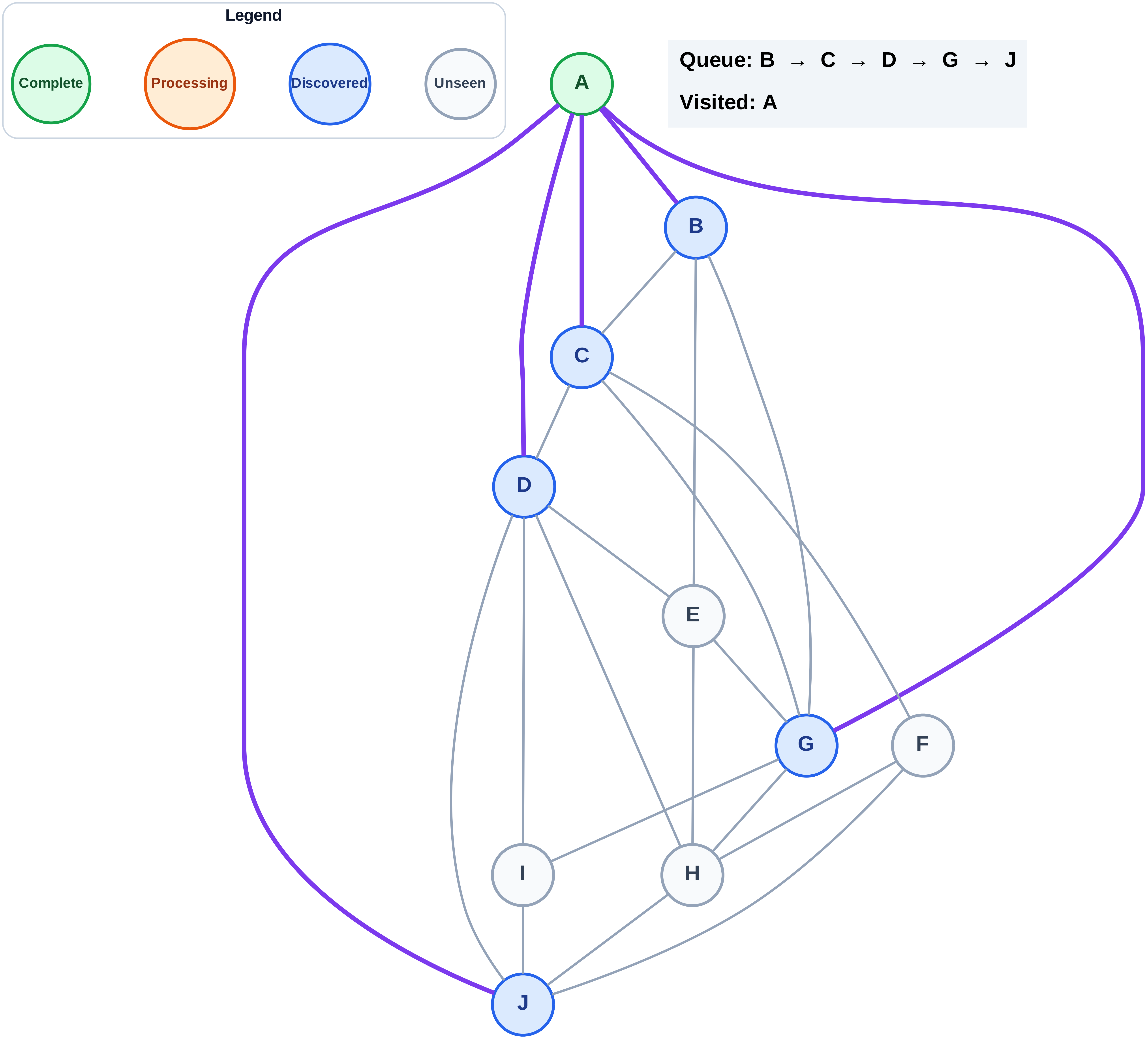
Step 1: Start at A




BFS Traversal
Step 2: Process node A
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: (none)
A
B
C
D
E
G
F
I
H
J

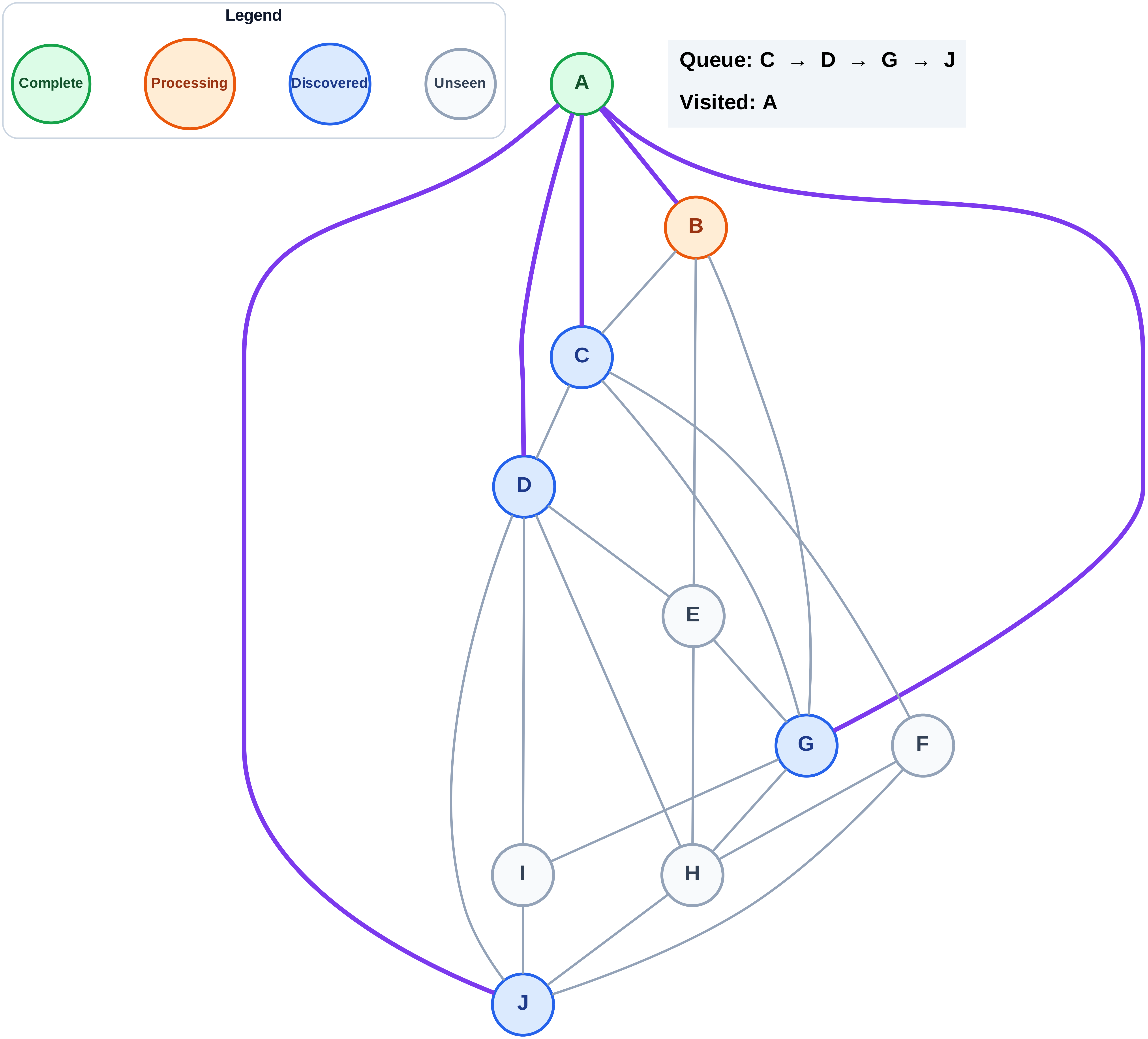
BFS Traversal

Step 3: Discover B, C, D, G, J from A



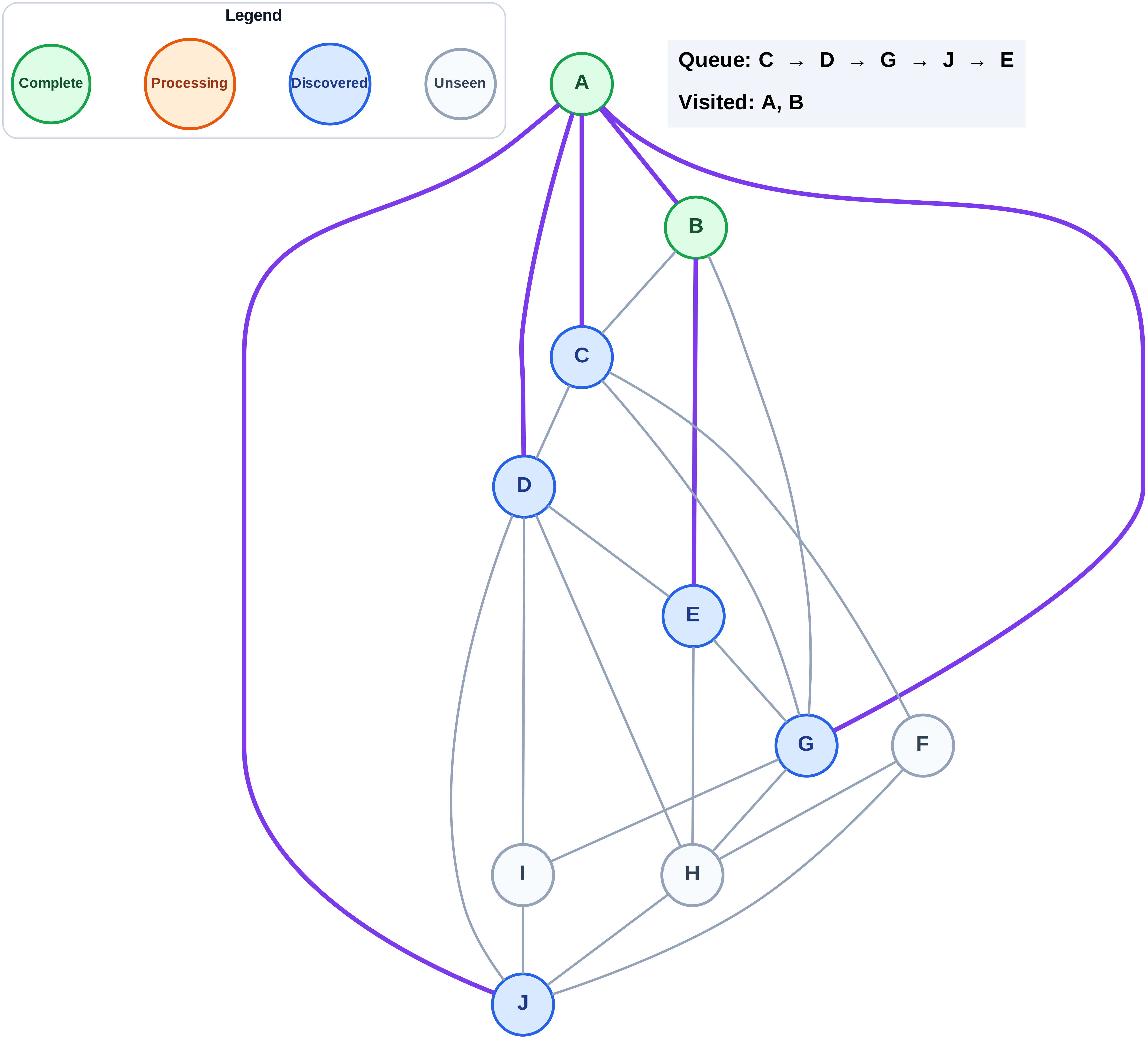
BFS Traversal

Step 4: Process node B



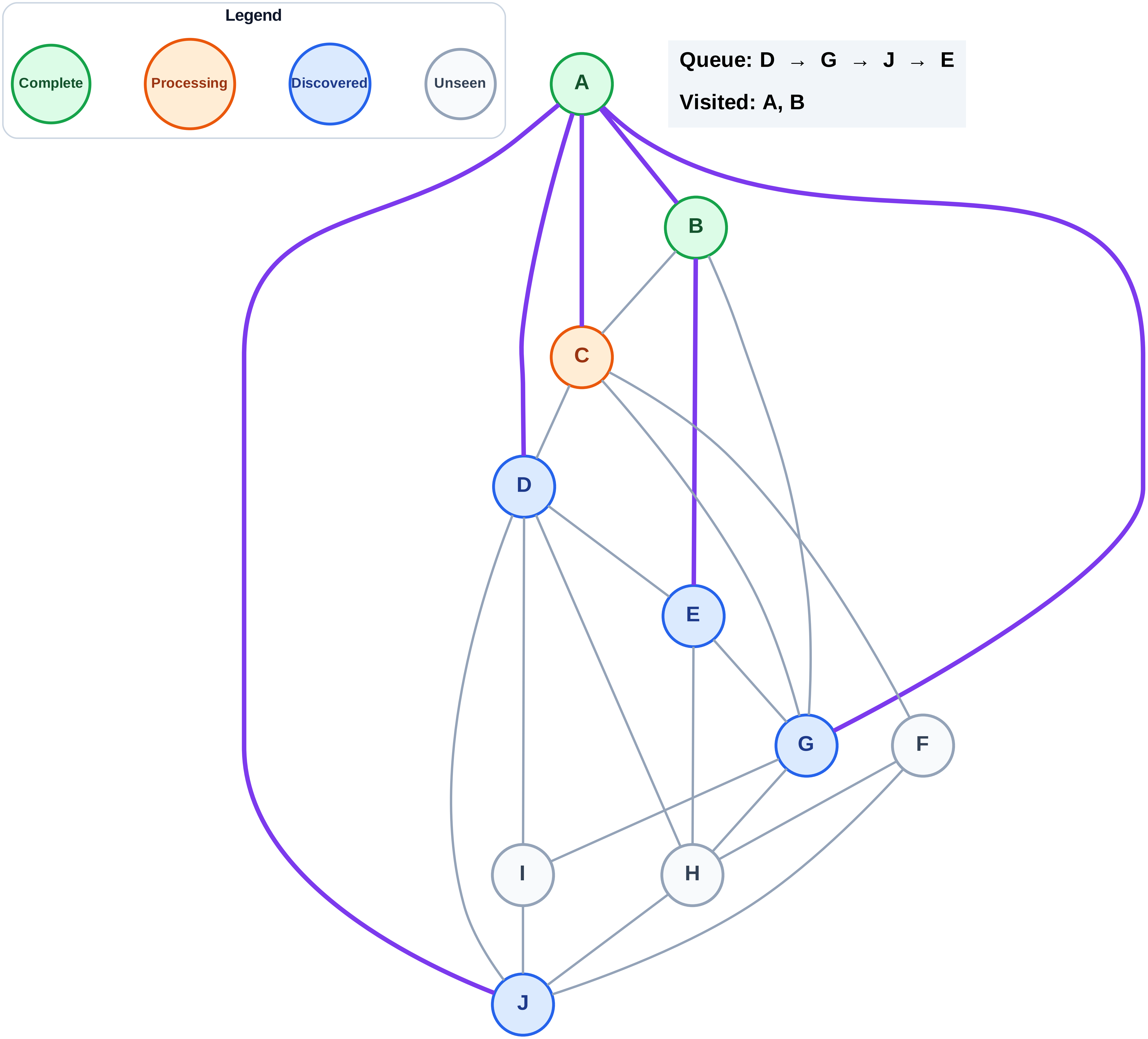
BFS Traversal

Step 5: Discover E from B



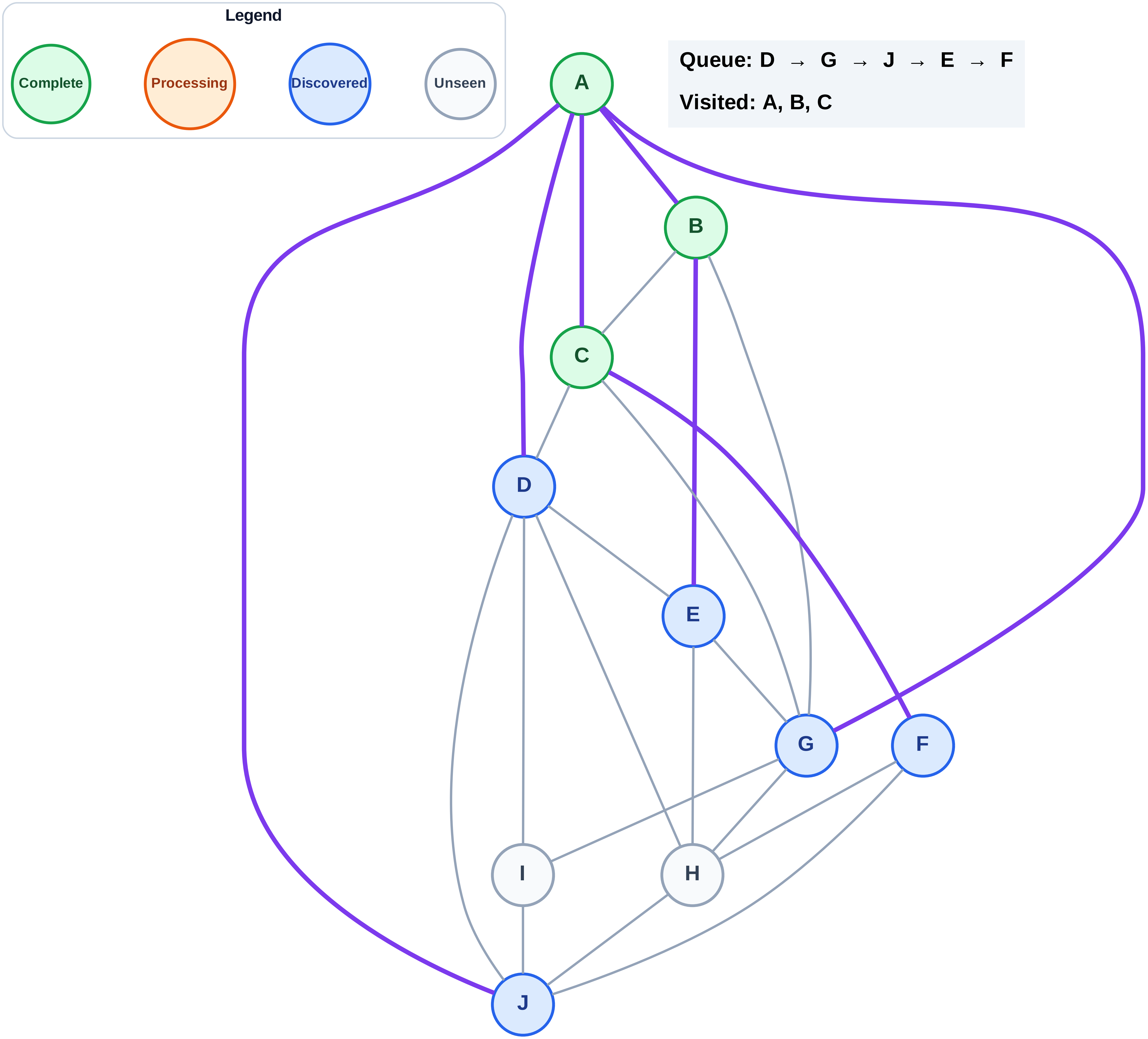
BFS Traversal

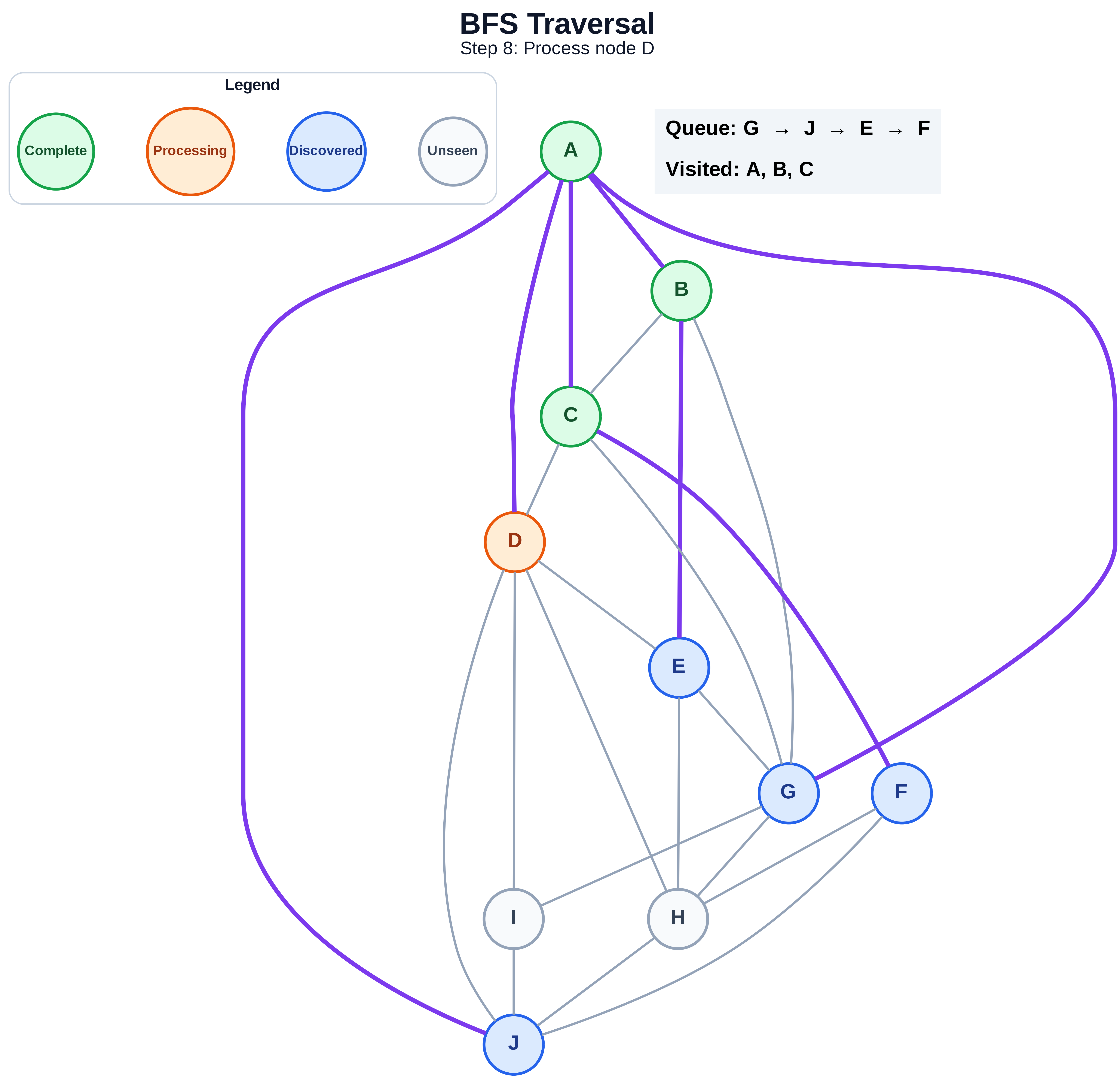
Step 6: Process node C



BFS Traversal

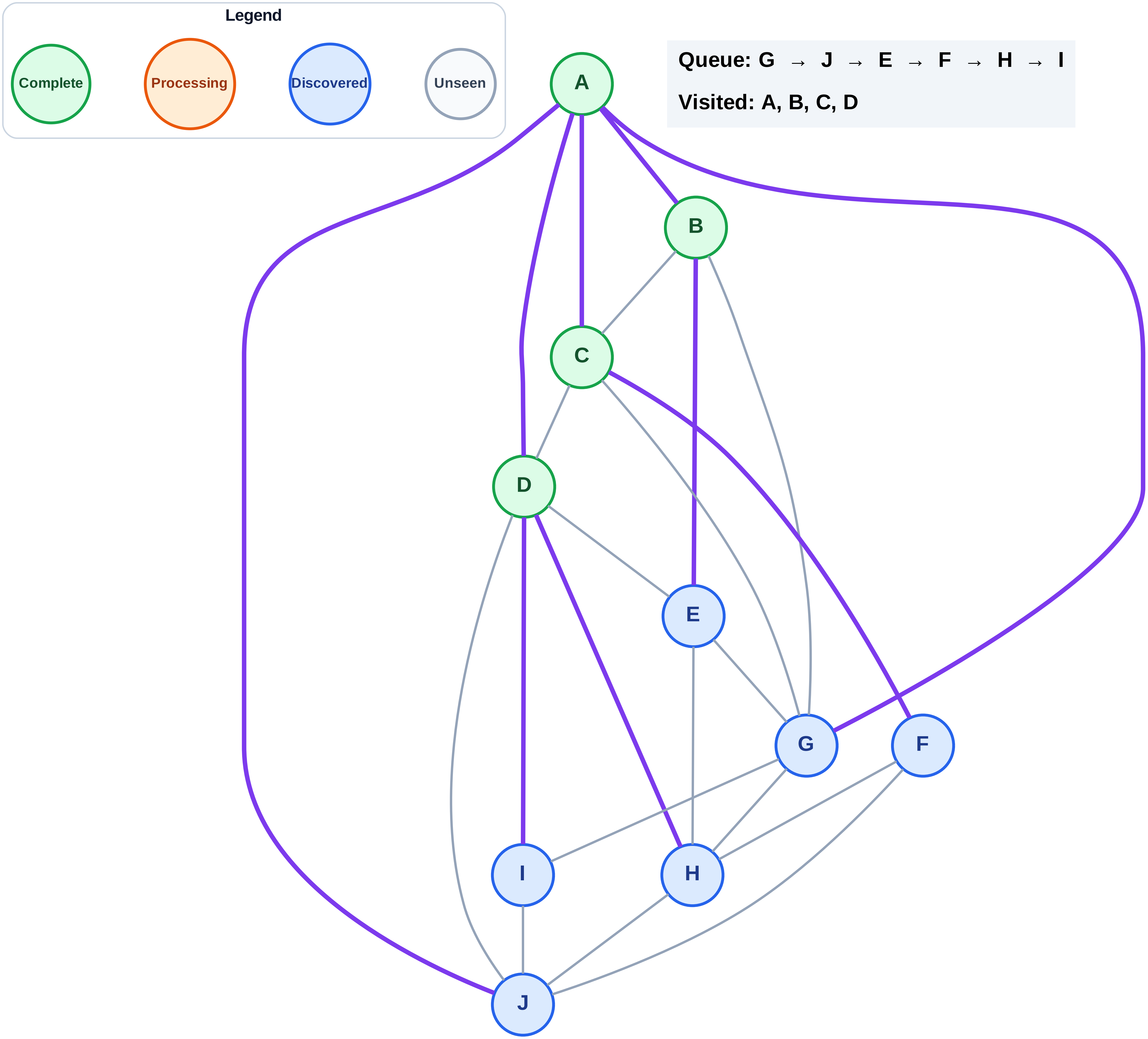
Step 7: Discover F from C




BFS Traversal
Step 8: Process node D
Legend
Complete
Processing
Discovered
Unseen
Queue: G → J → E → F
Visited: A, B, C
A
B
C
D
E
G
F
I
H
J

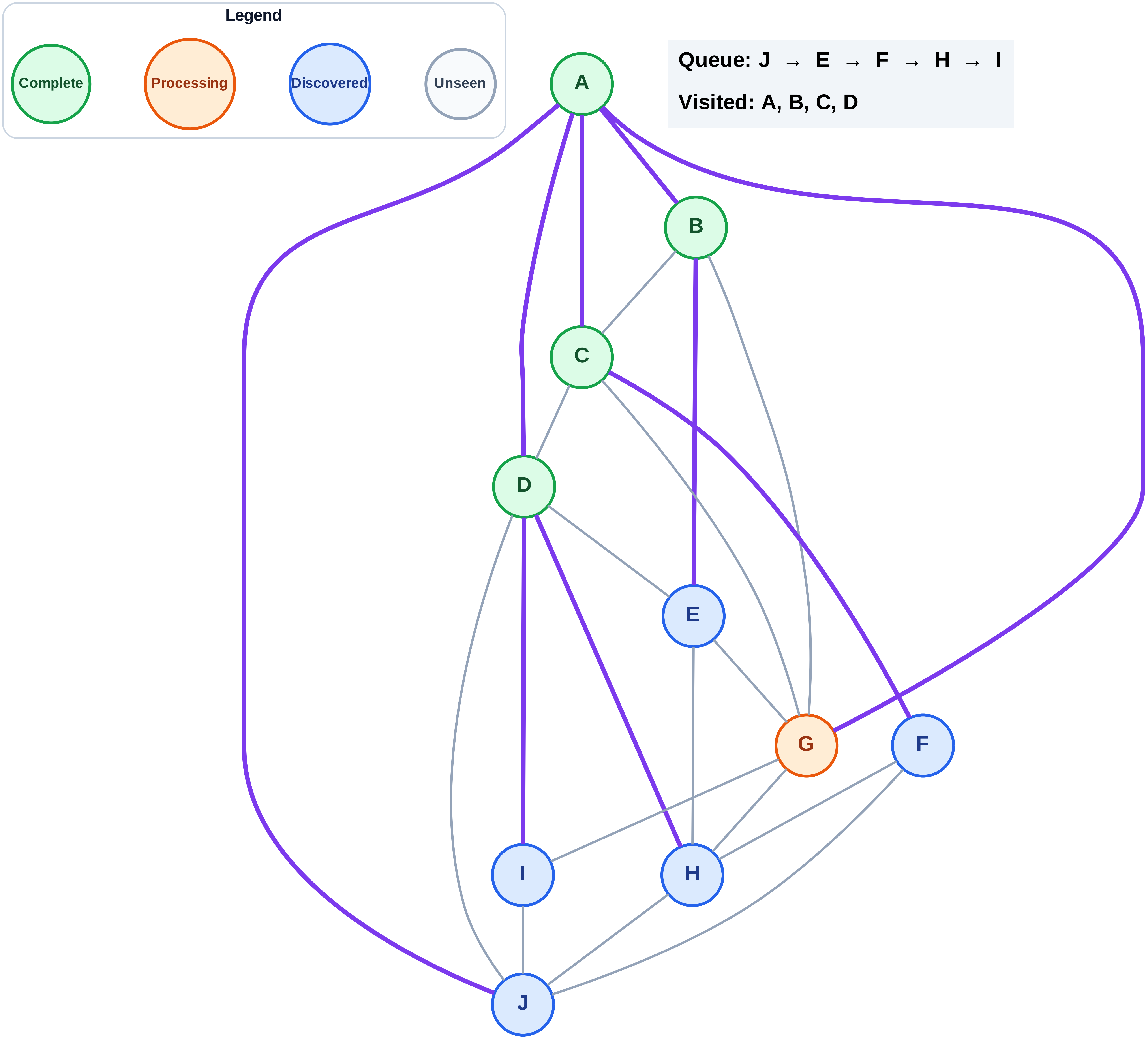
BFS Traversal

Step 9: Discover H, I from D



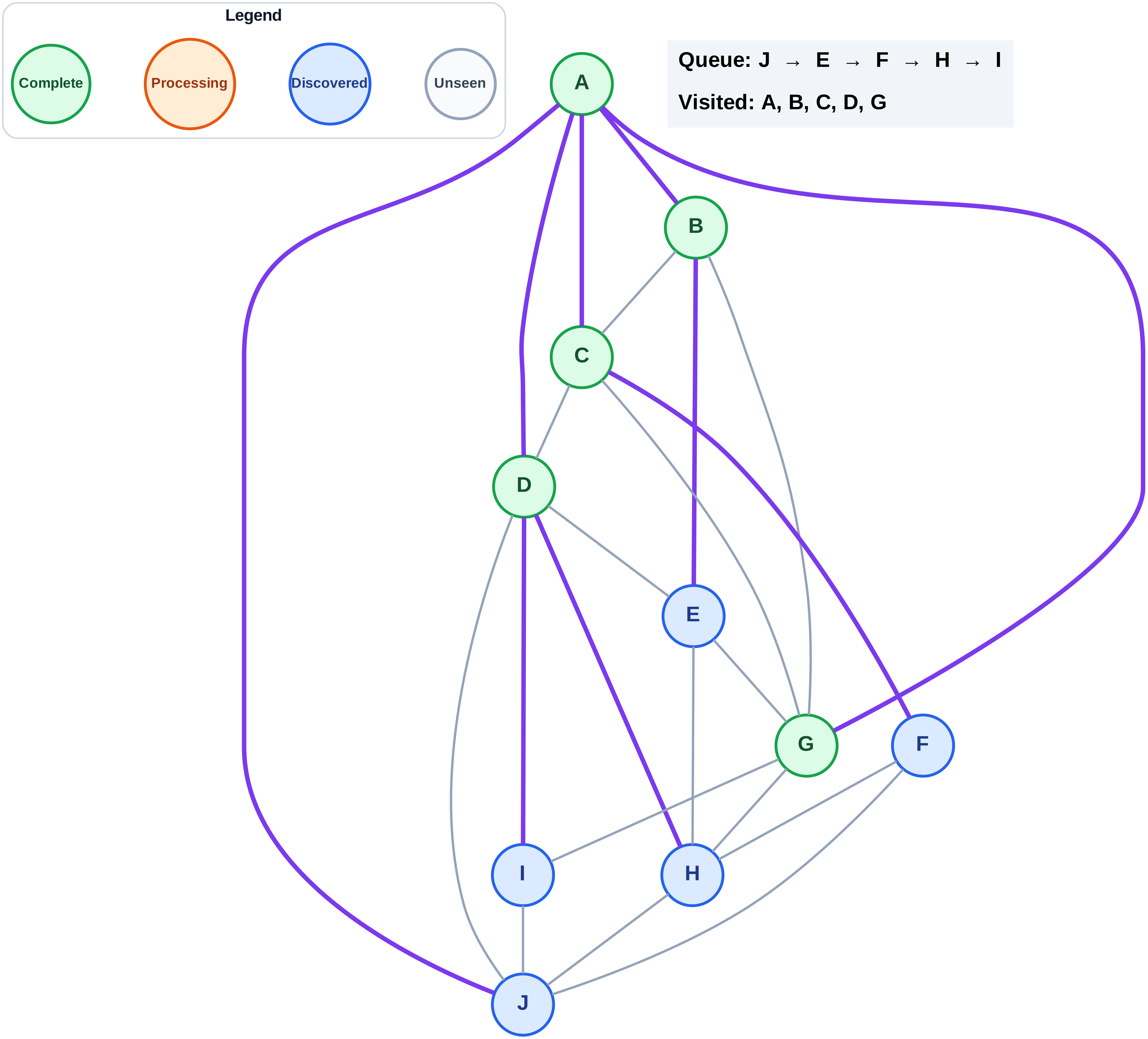
BFS Traversal

Step 10: Process node G



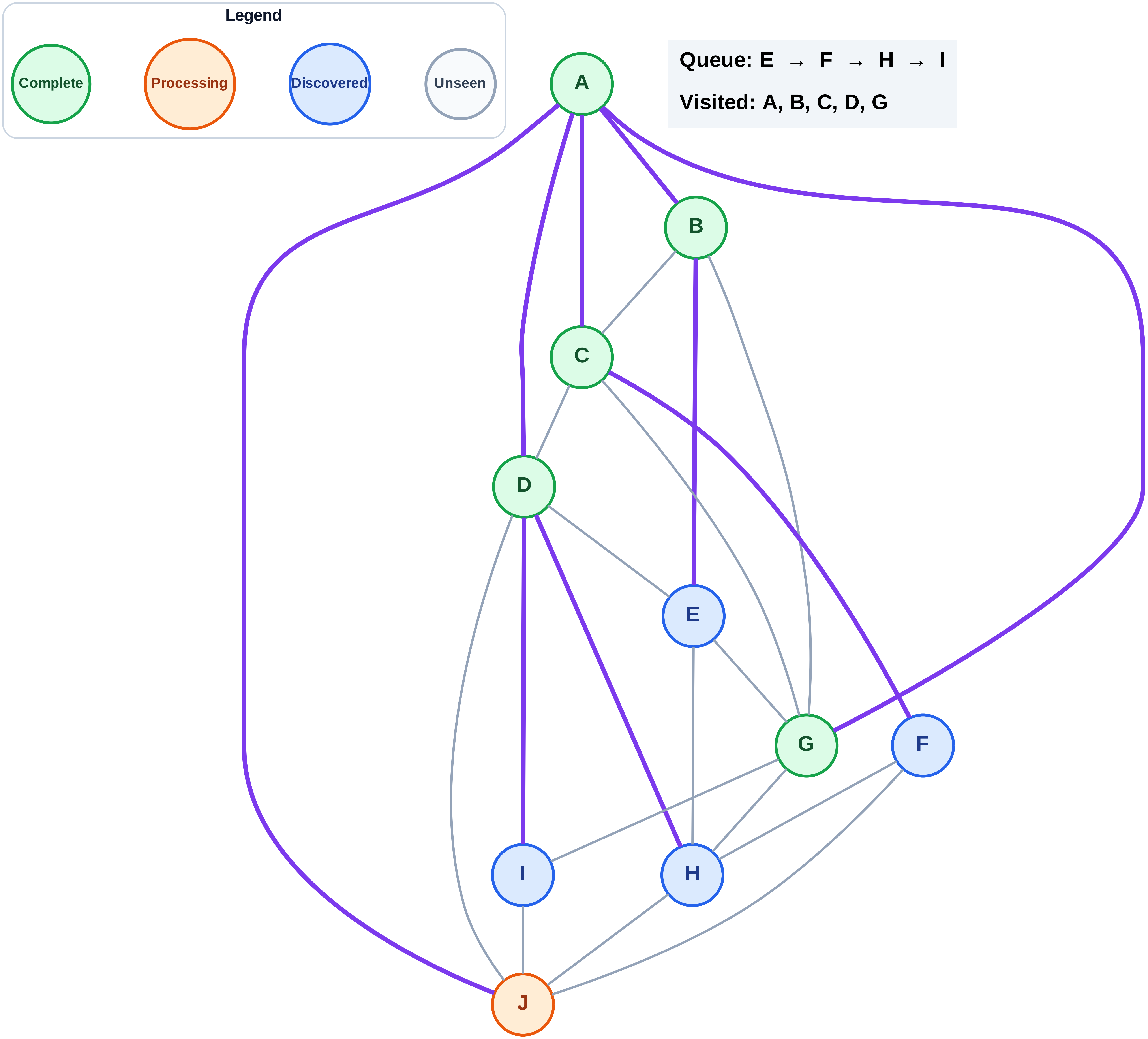
BFS Traversal

Step 11: No new neighbors from G



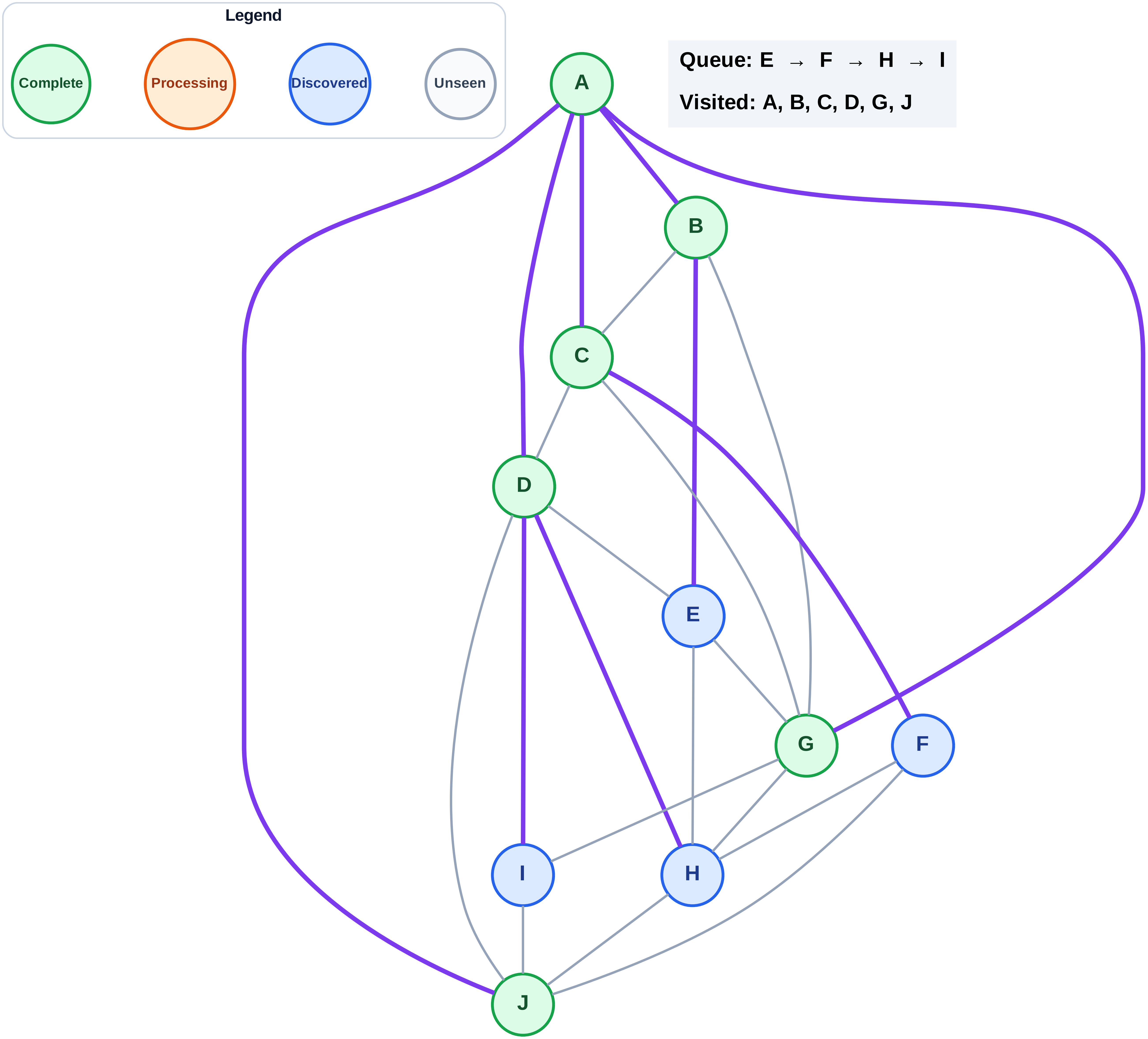
BFS Traversal

Step 12: Process node J



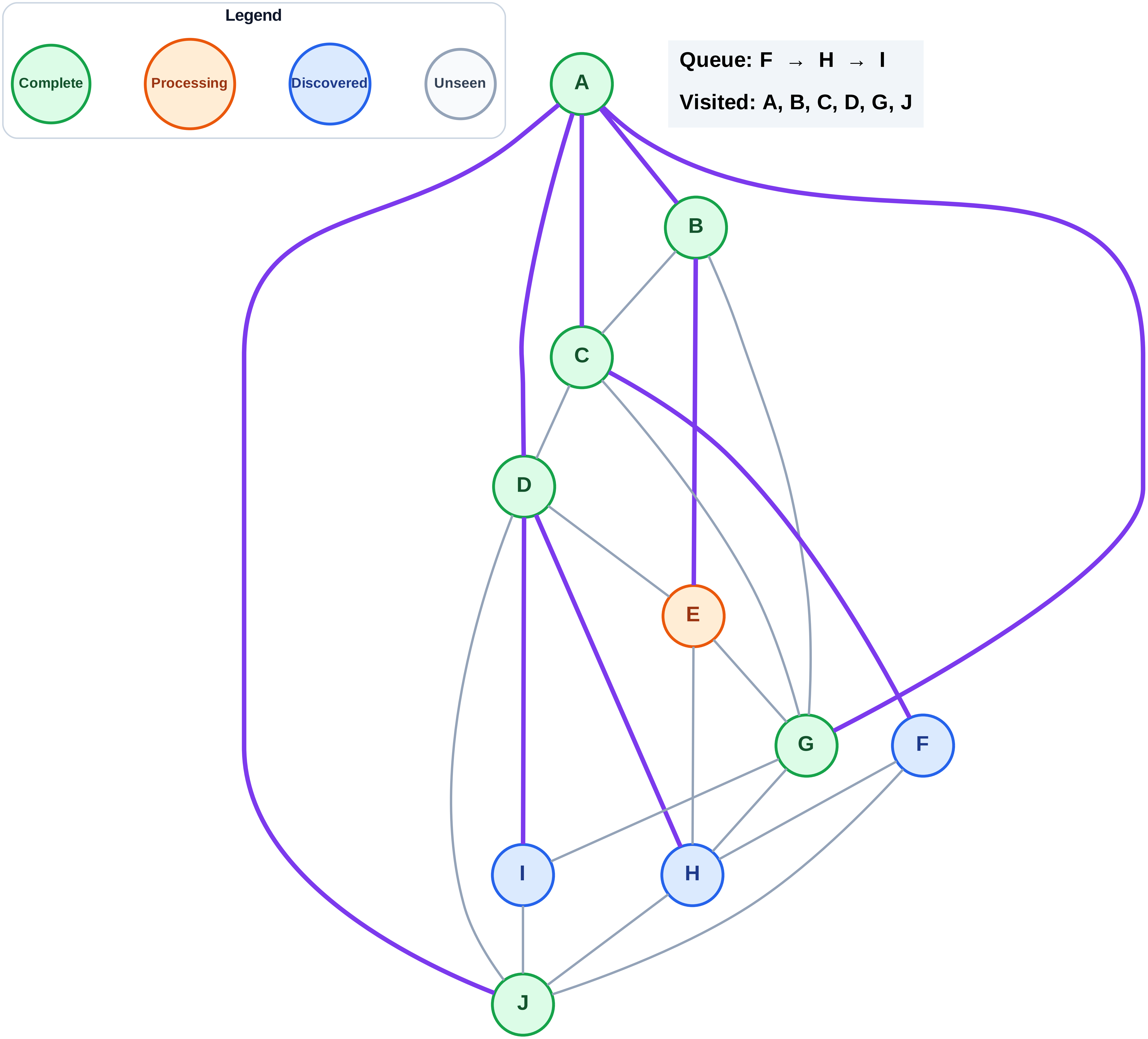
BFS Traversal

Step 13: No new neighbors from J



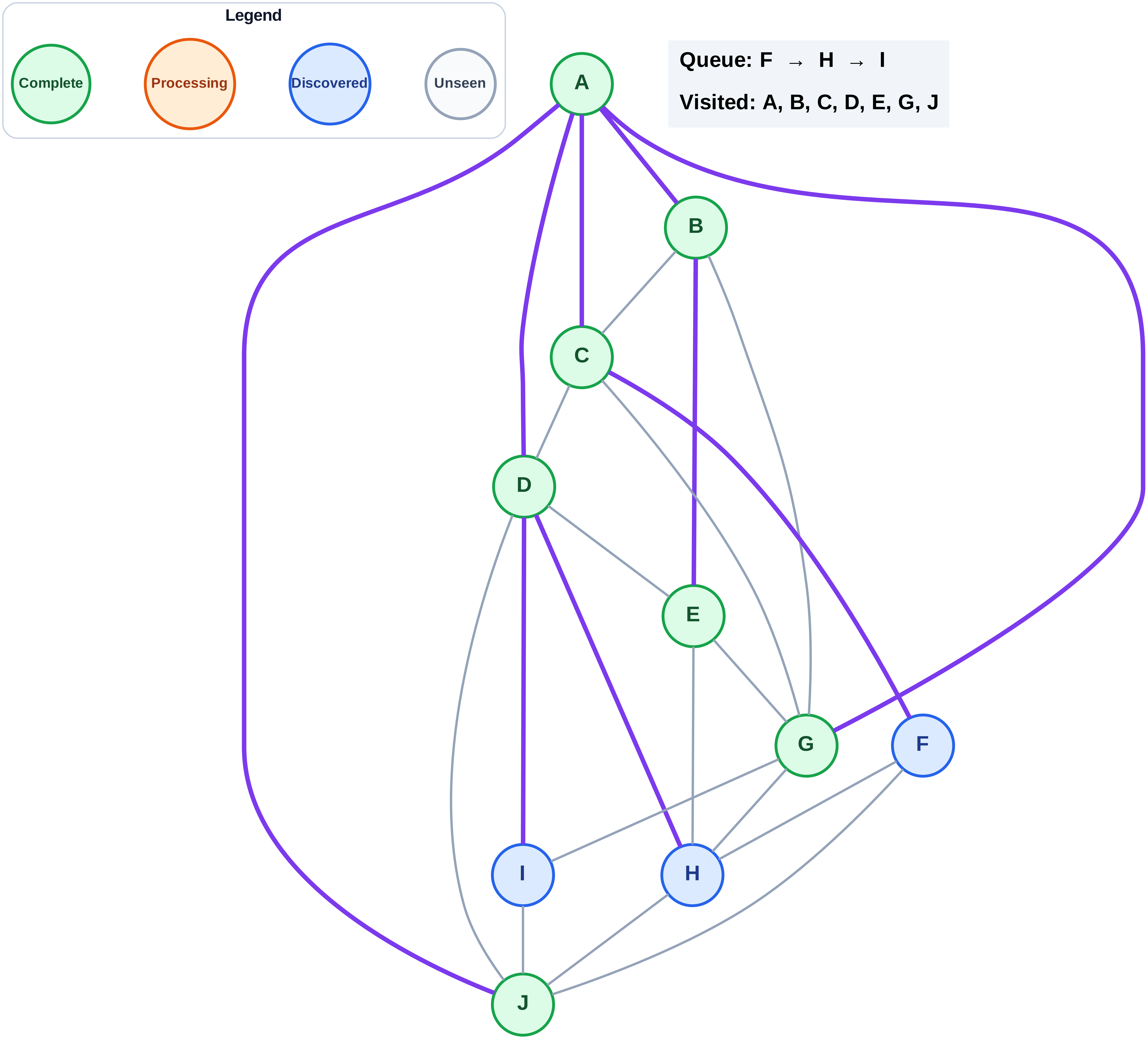
BFS Traversal

Step 14: Process node E



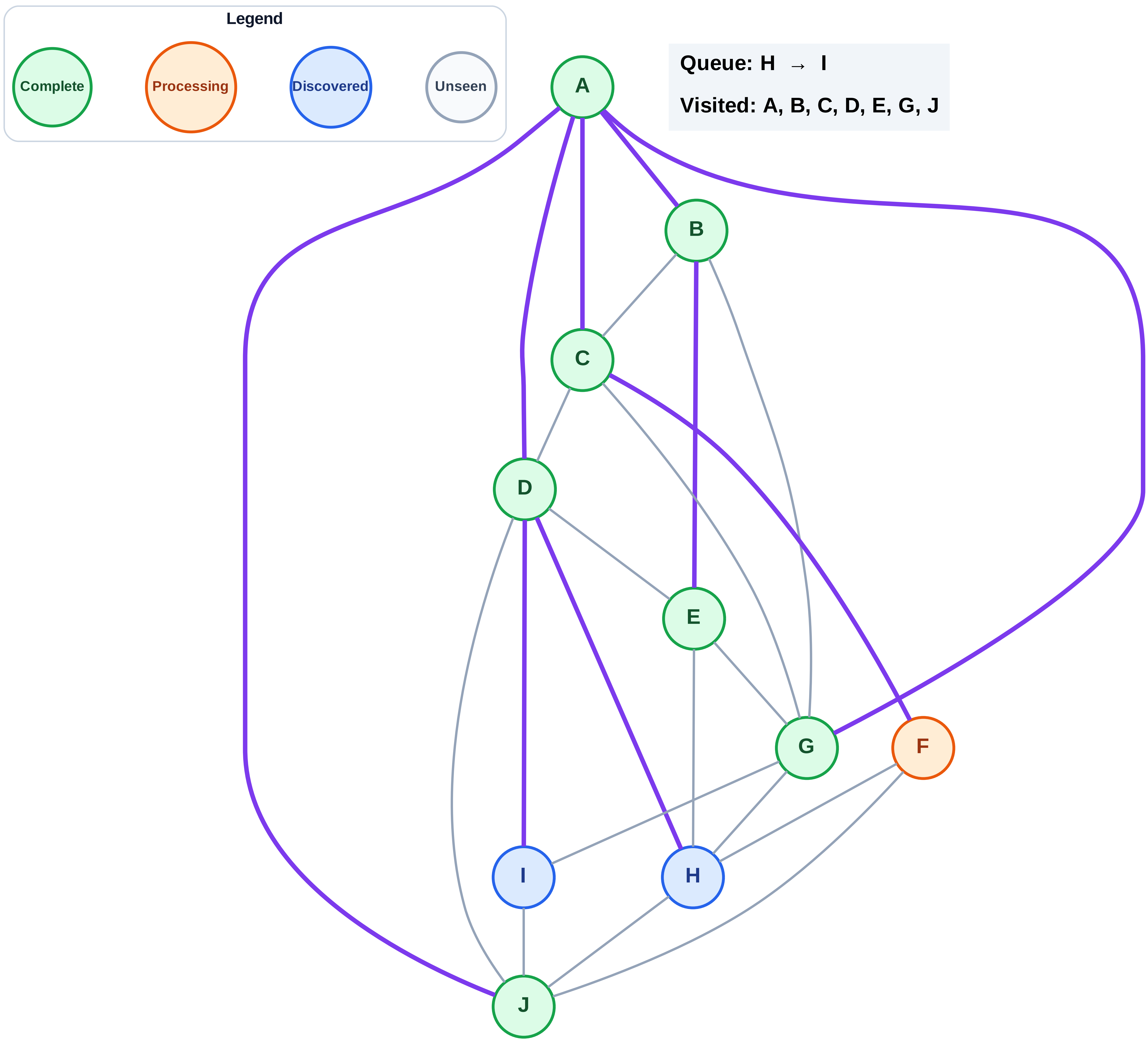
BFS Traversal

Step 15: No new neighbors from E



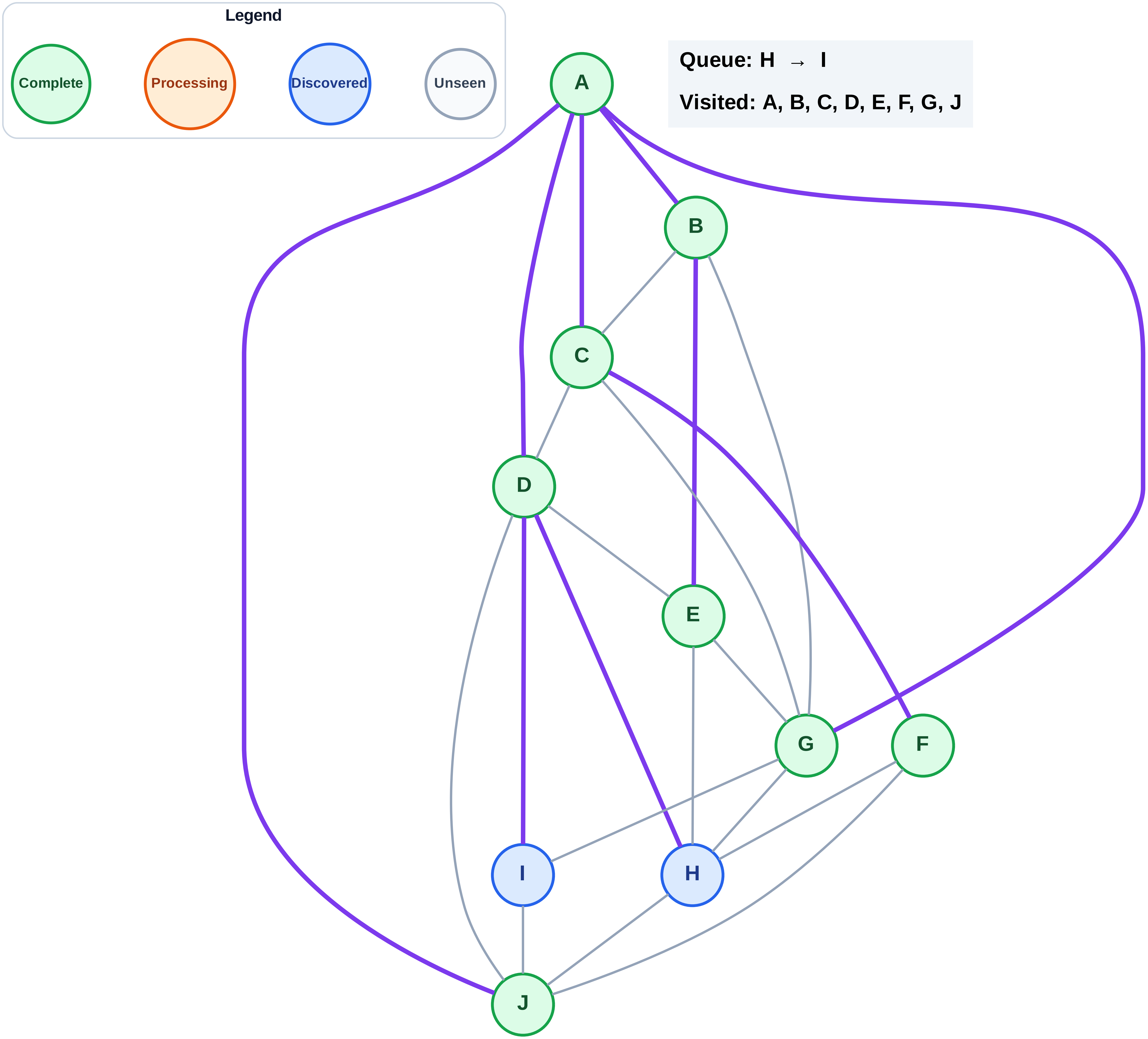
BFS Traversal

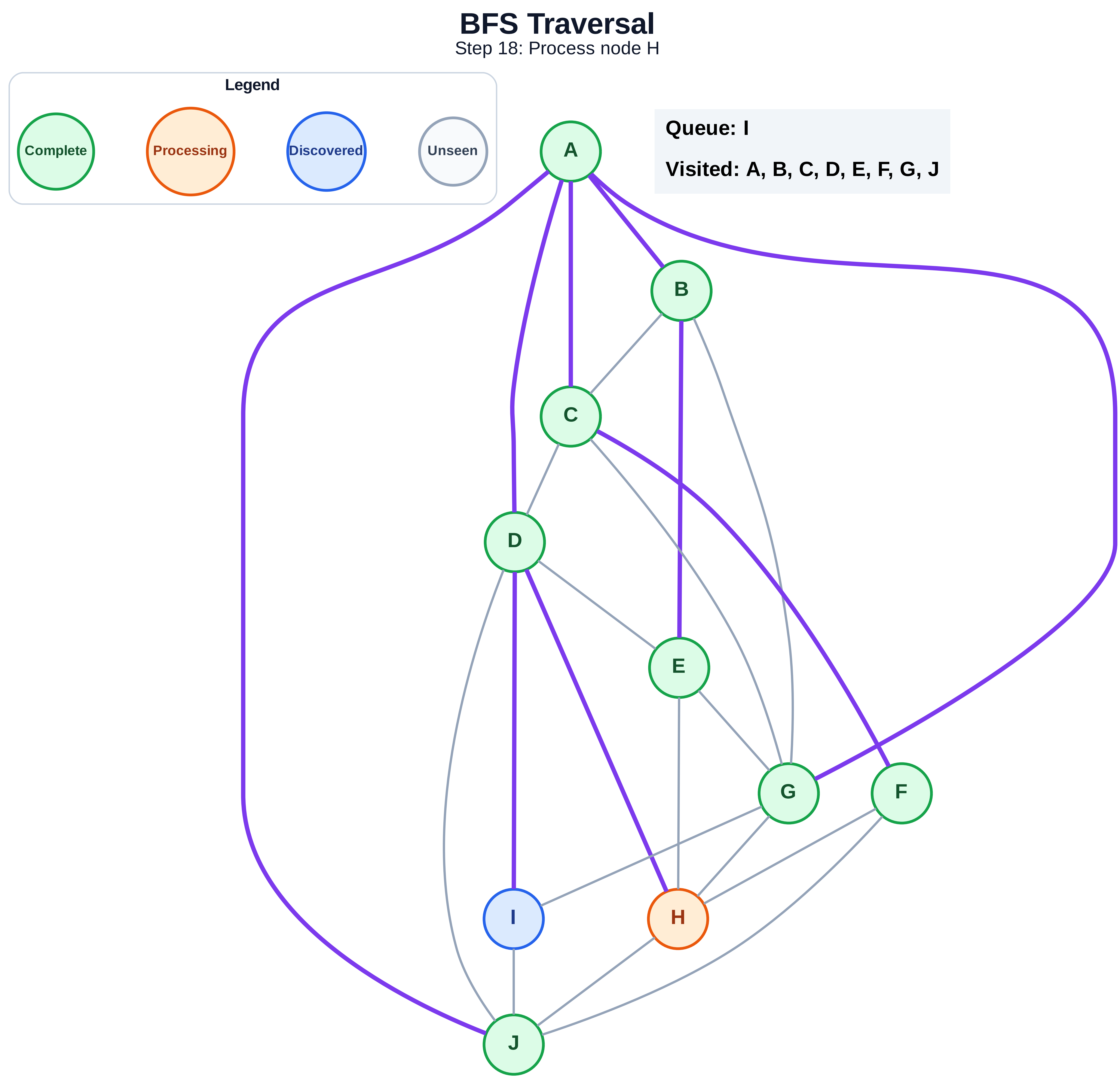
Step 16: Process node F



BFS Traversal

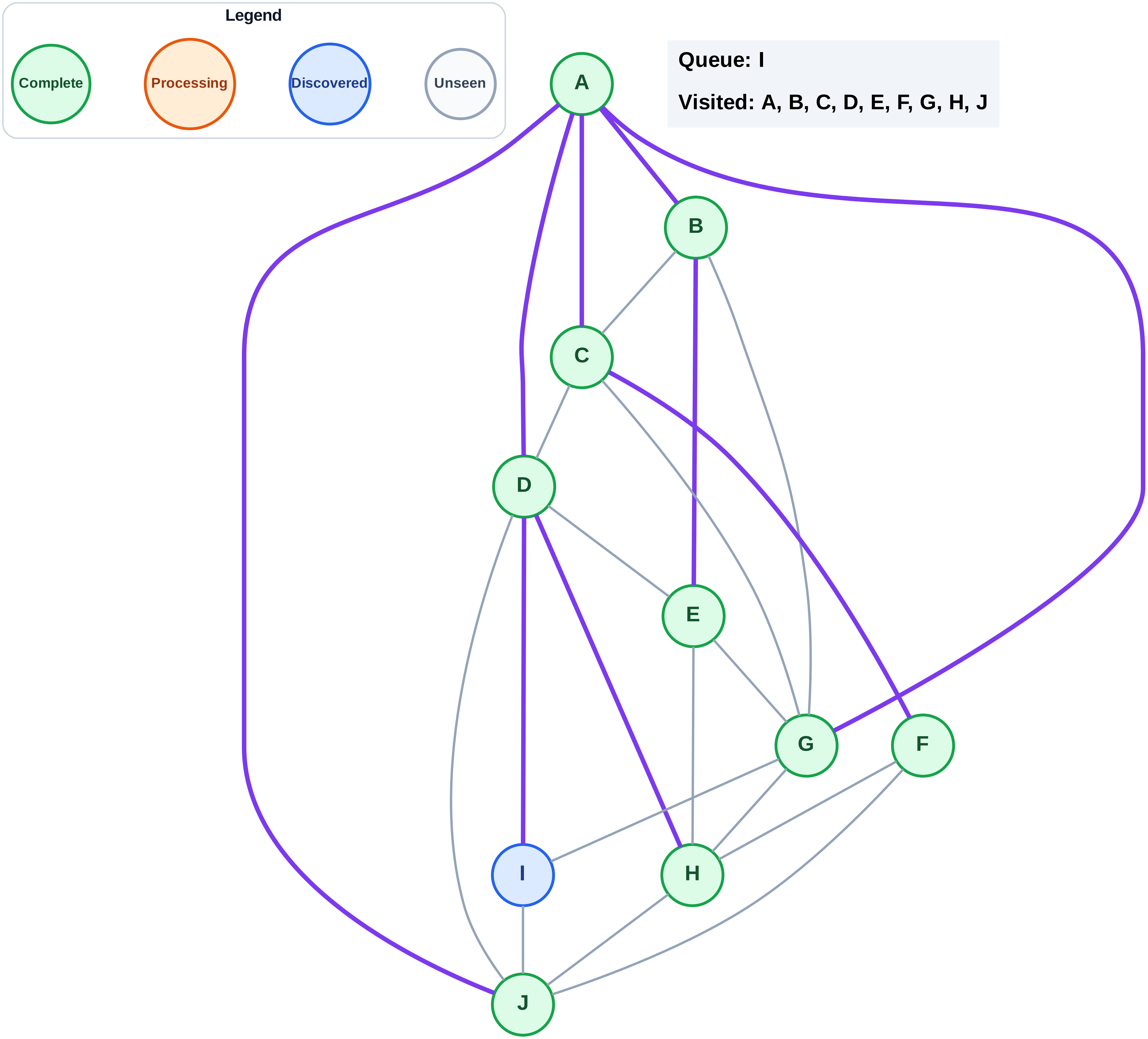
Step 17: No new neighbors from F




BFS Traversal
Step 18: Process node H
Legend
Complete
Processing
Discovered
Unseen
Queue: I
Visited: A, B, C, D, E, F, G, J
A
B
C
D
E
F
G
H
I
J

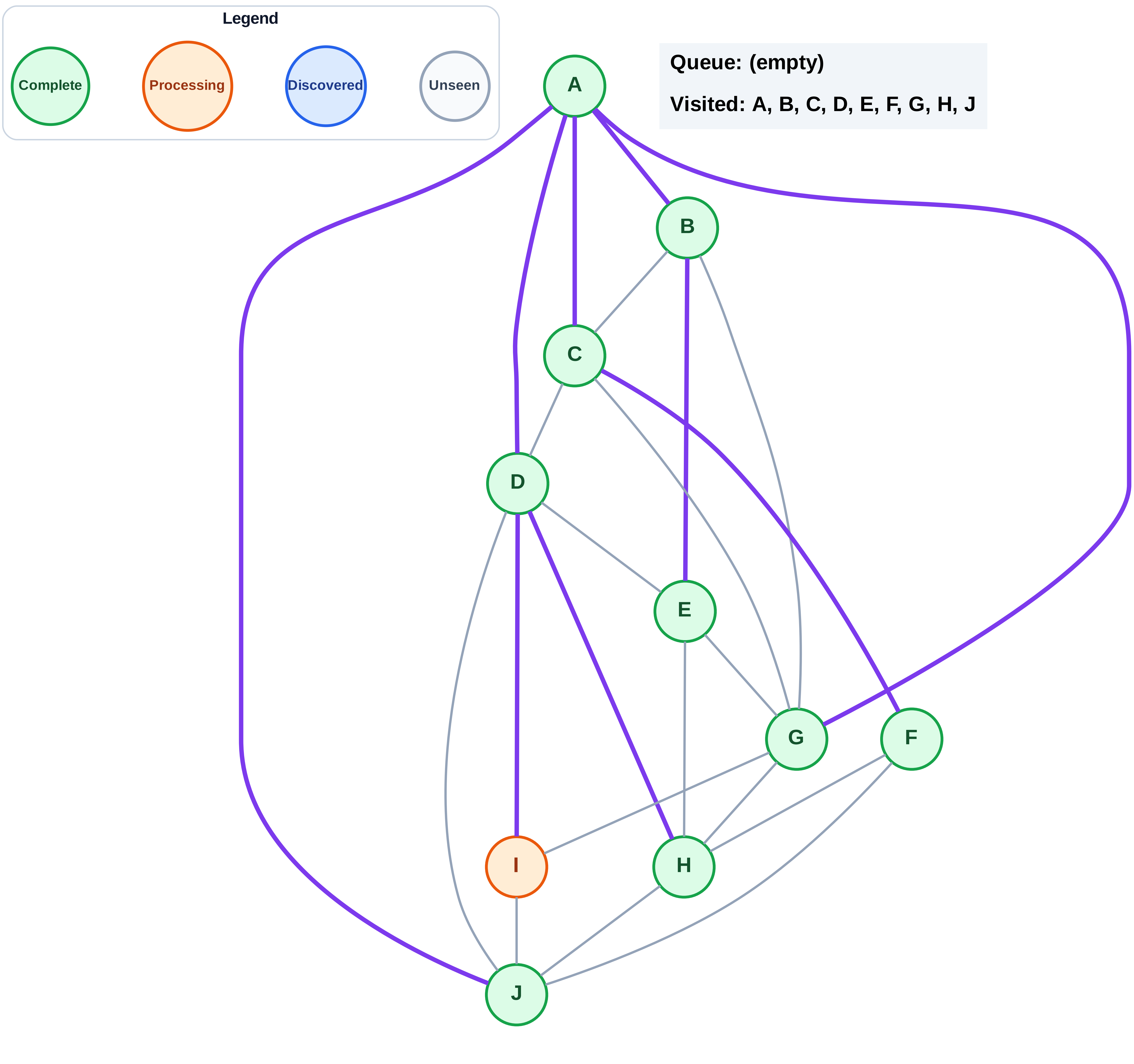
BFS Traversal

Step 19: No new neighbors from H



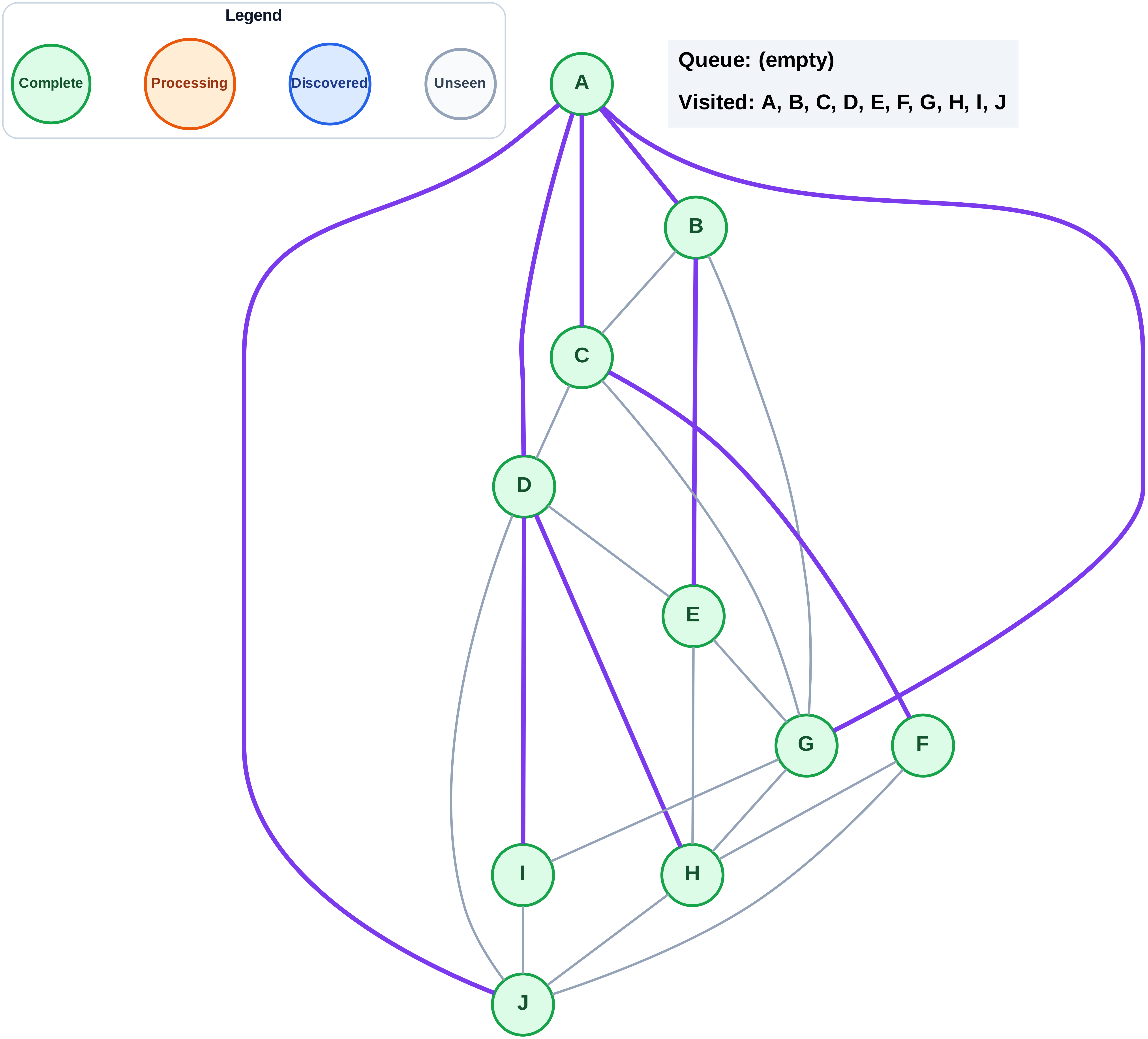
BFS Traversal

Step 20: Process node I



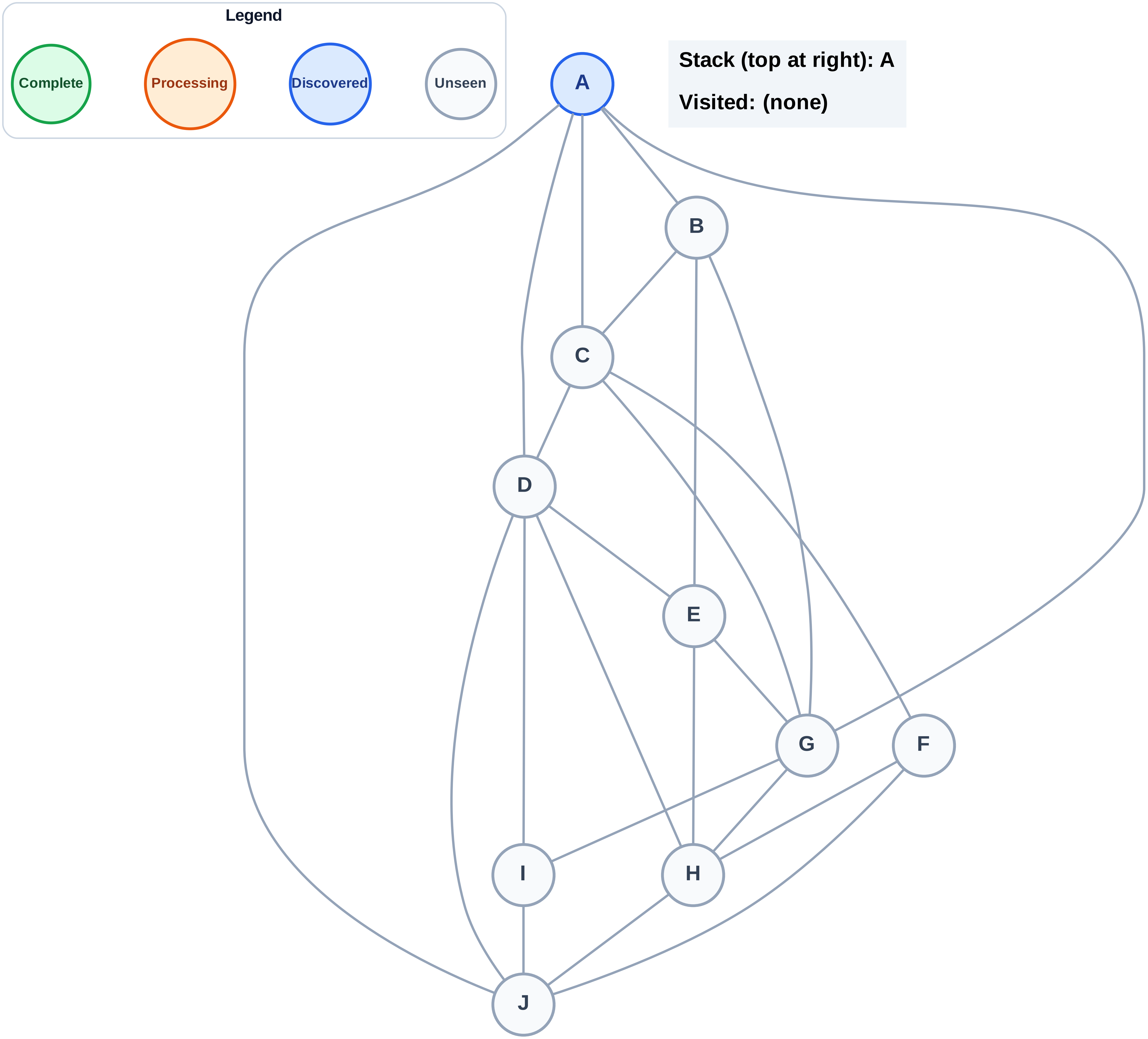
BFS Traversal

Step 21: No new neighbors from I



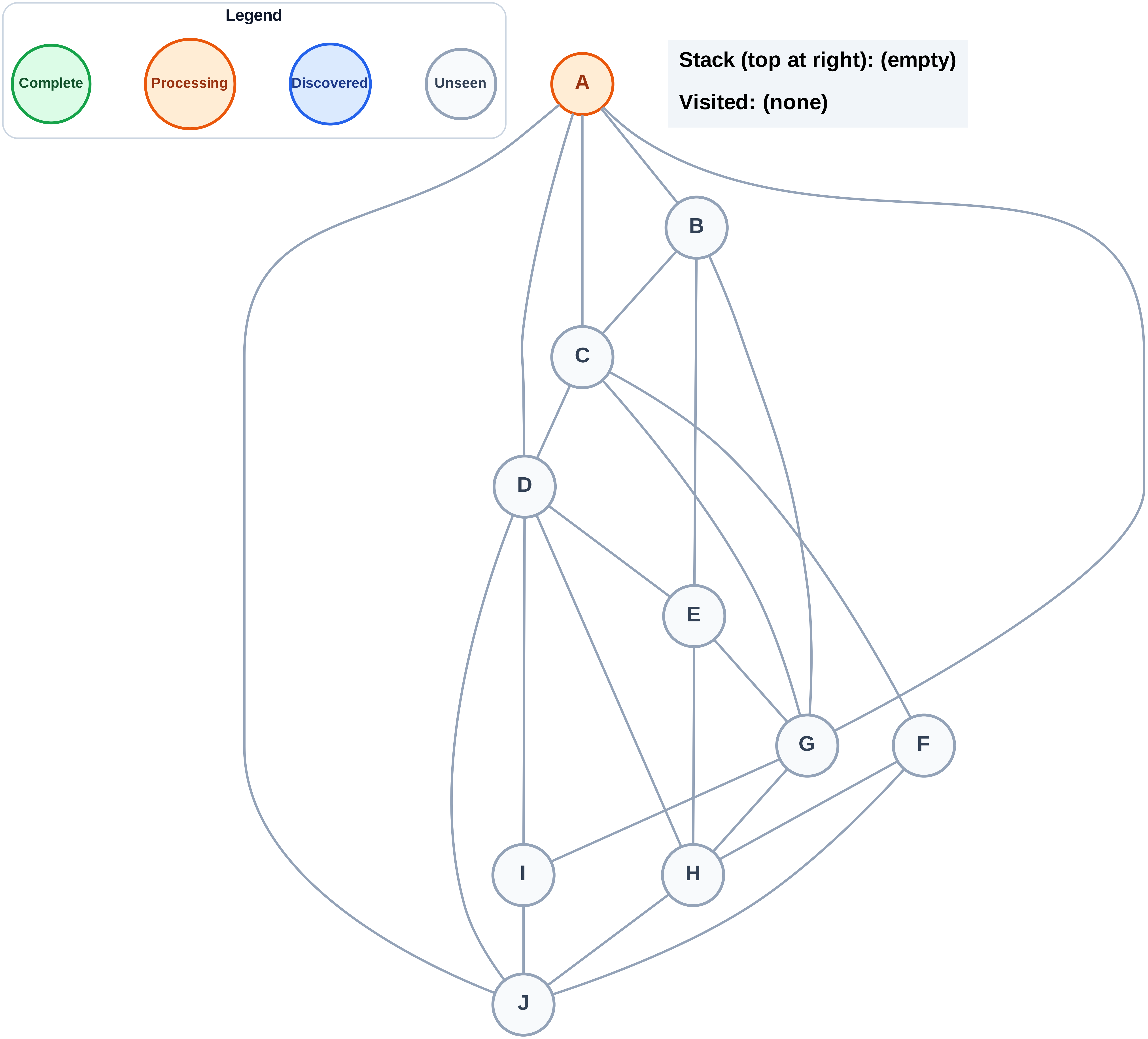
DFS Traversal

Step 1: Start at A



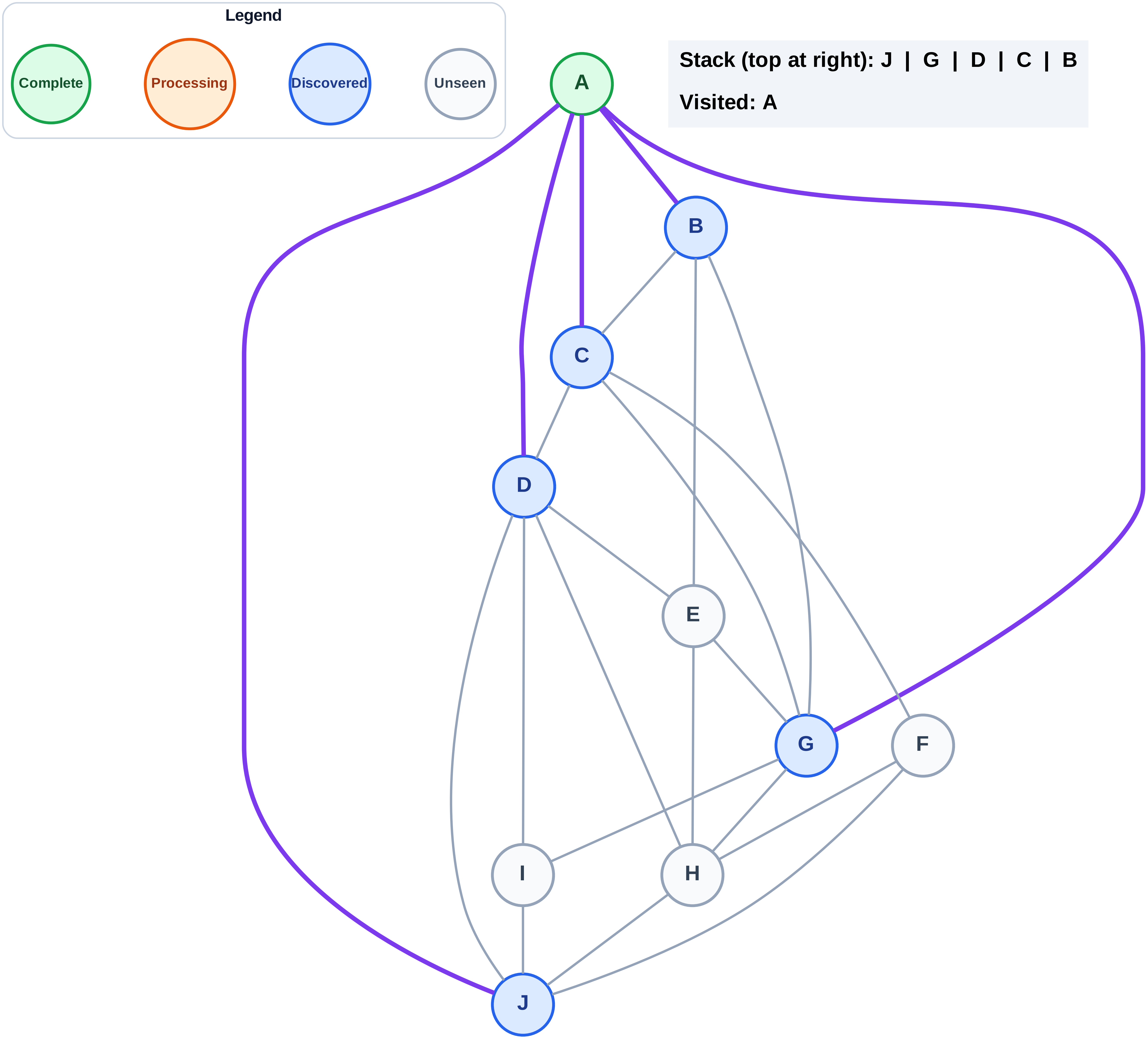
DFS Traversal

Step 2: Process node A



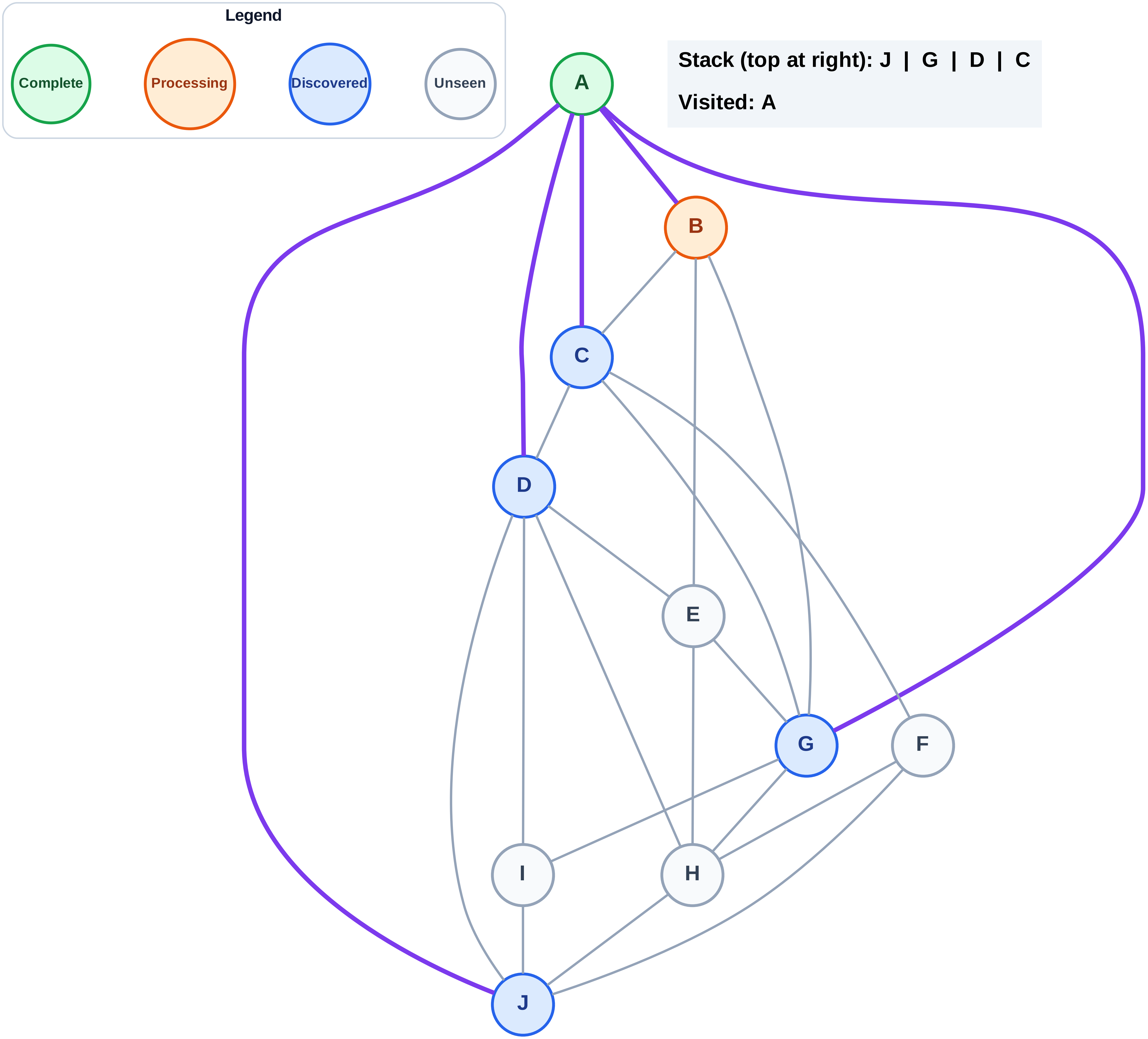
DFS Traversal

Step 3: Discover J, G, D, C, B from A



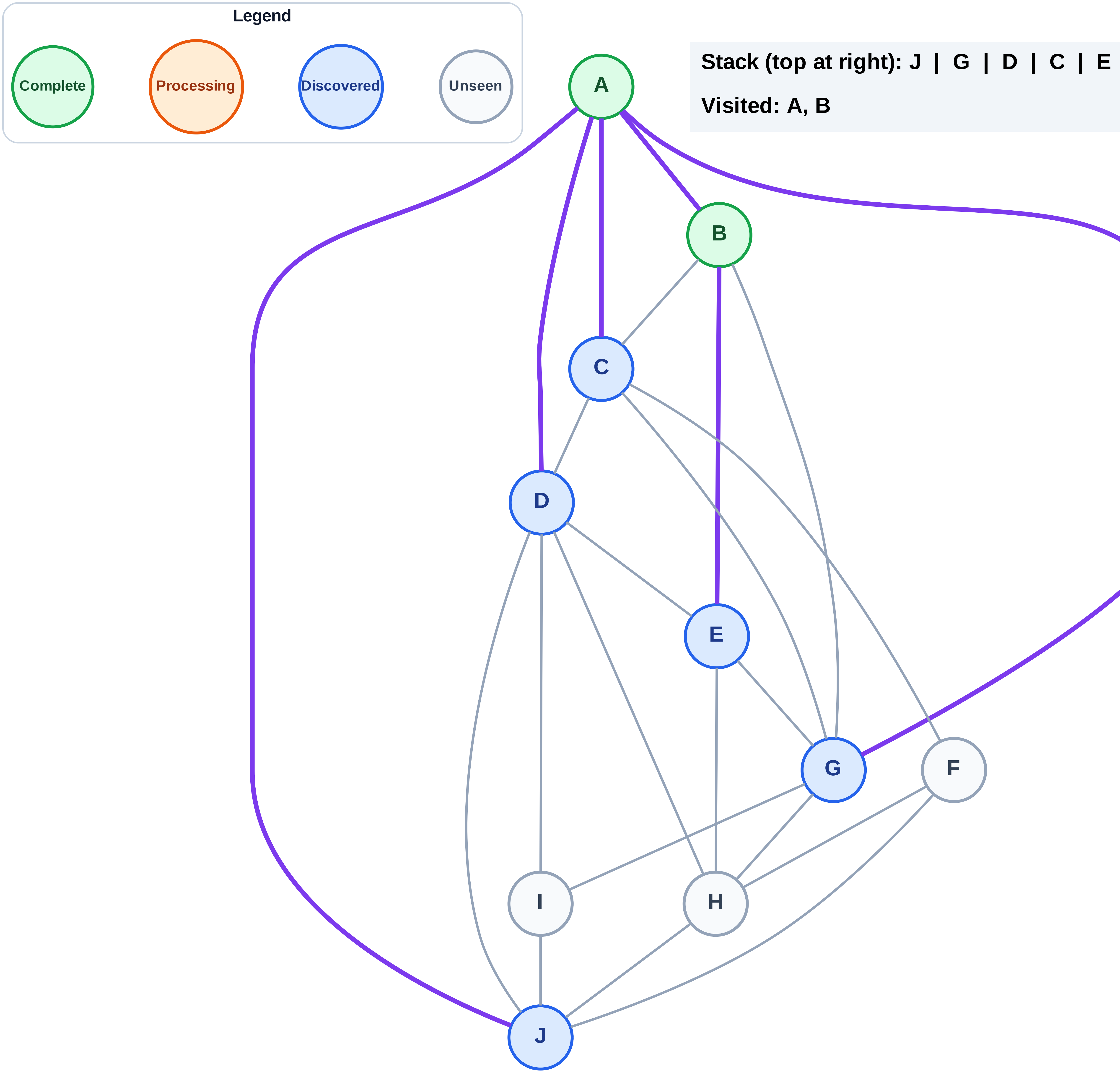
DFS Traversal

Step 4: Process node B



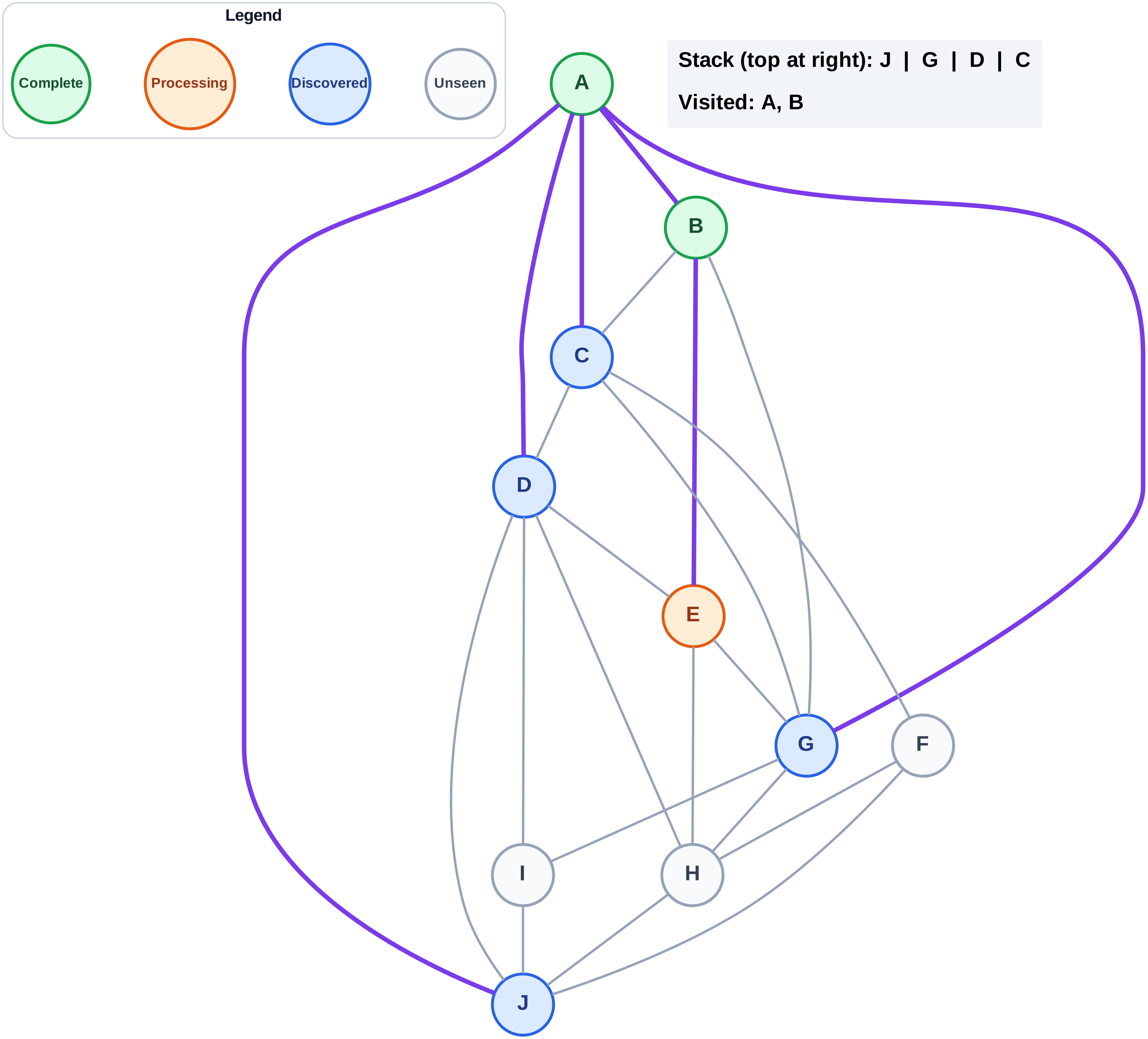
DFS Traversal

Step 5: Discover E from B



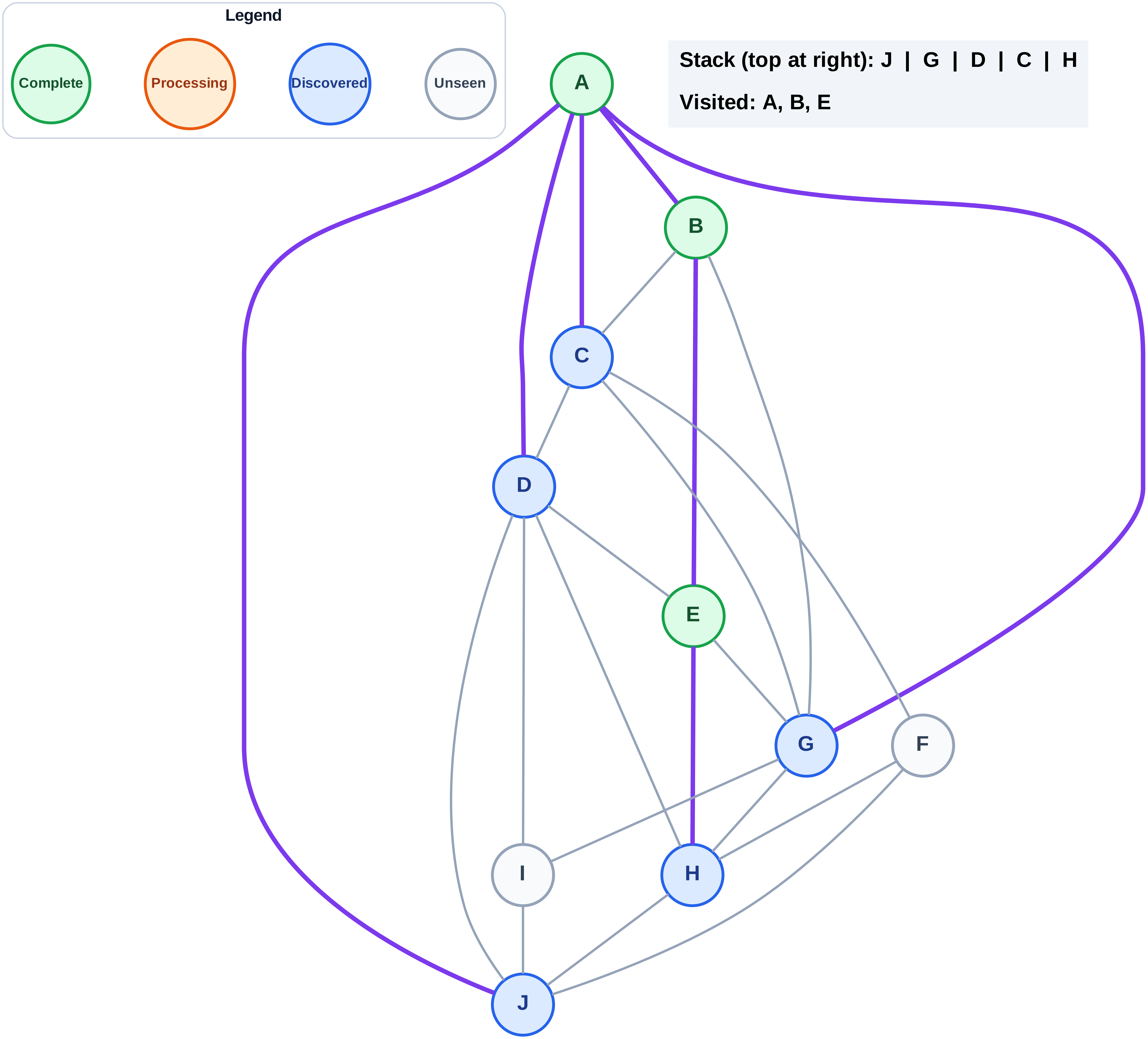
DFS Traversal

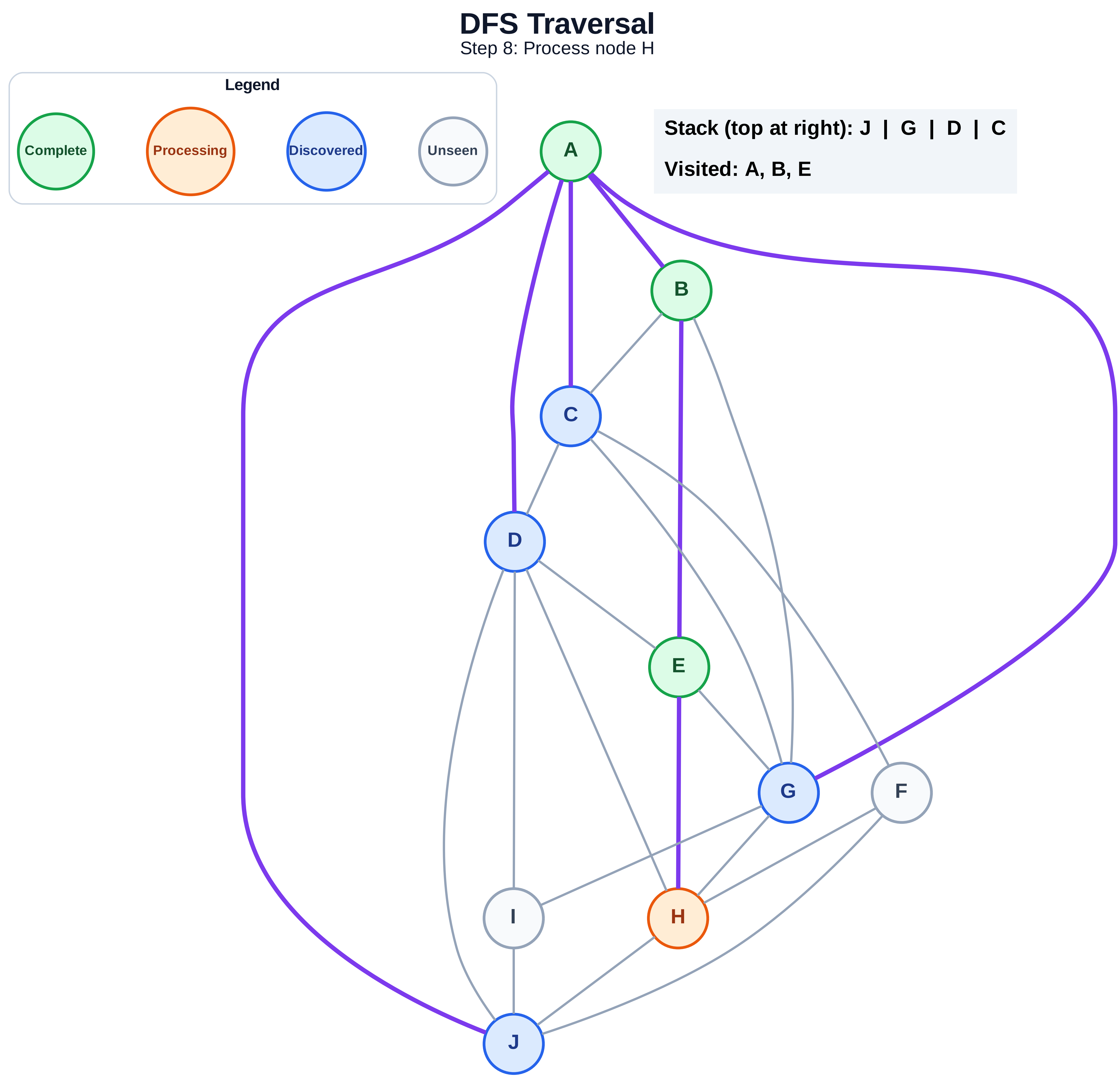
Step 6: Process node E



DFS Traversal

Step 7: Discover H from E

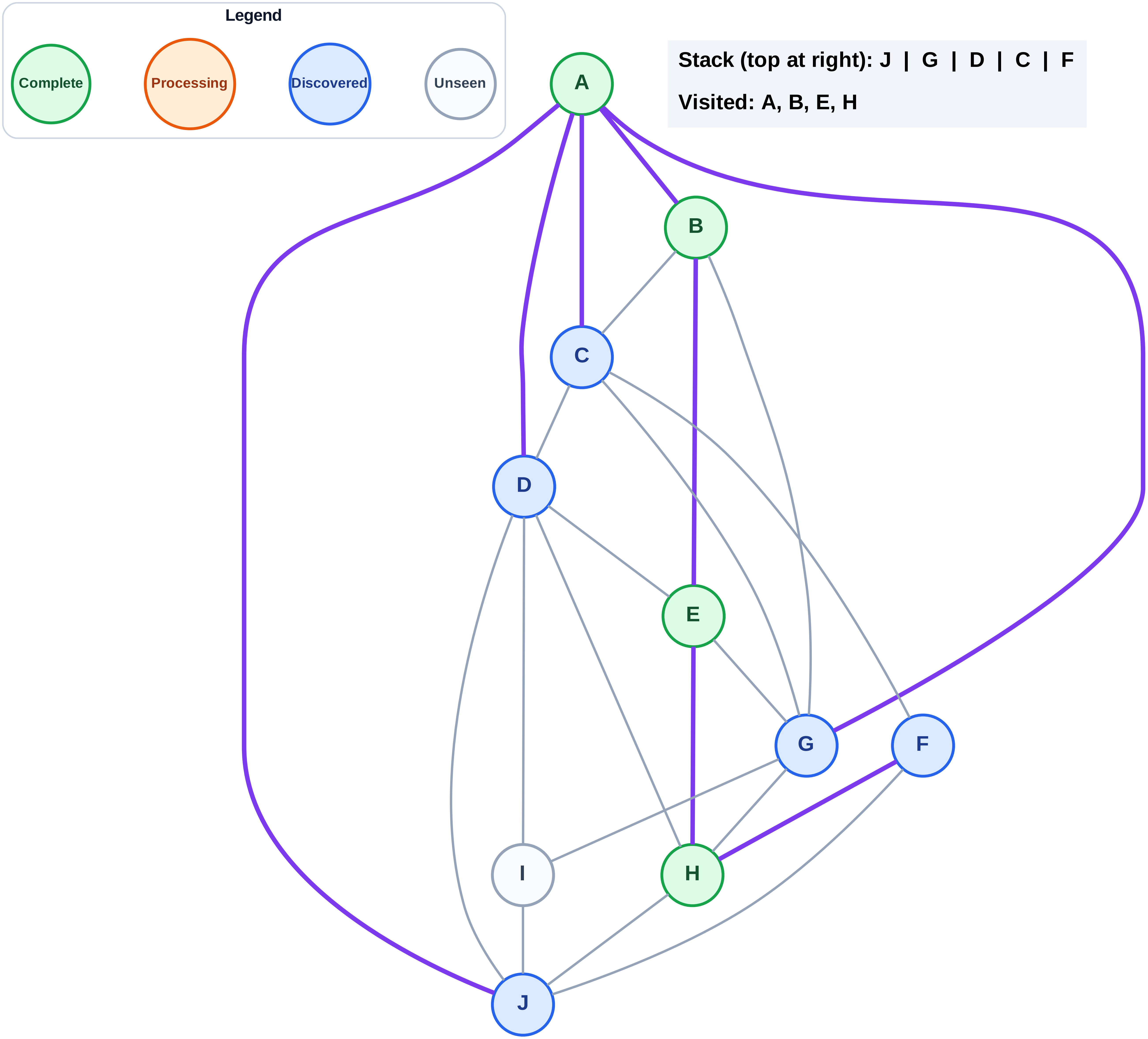




DFS Traversal
Step 8: Process node H
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G | D | C
Visited: A, B, E
A
B
C
D
E
F
G
H
I
J

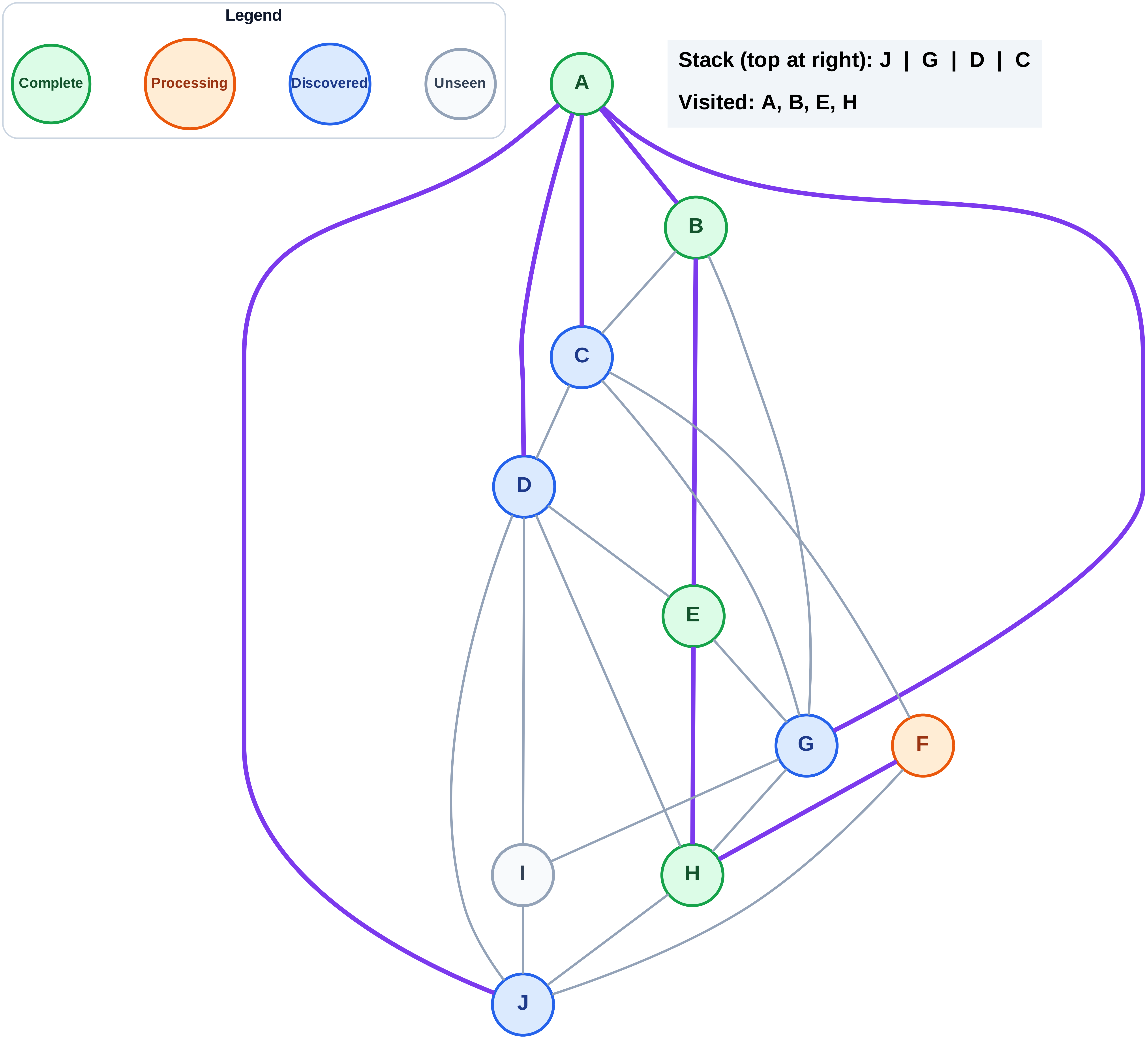
DFS Traversal

Step 9: Discover F from H



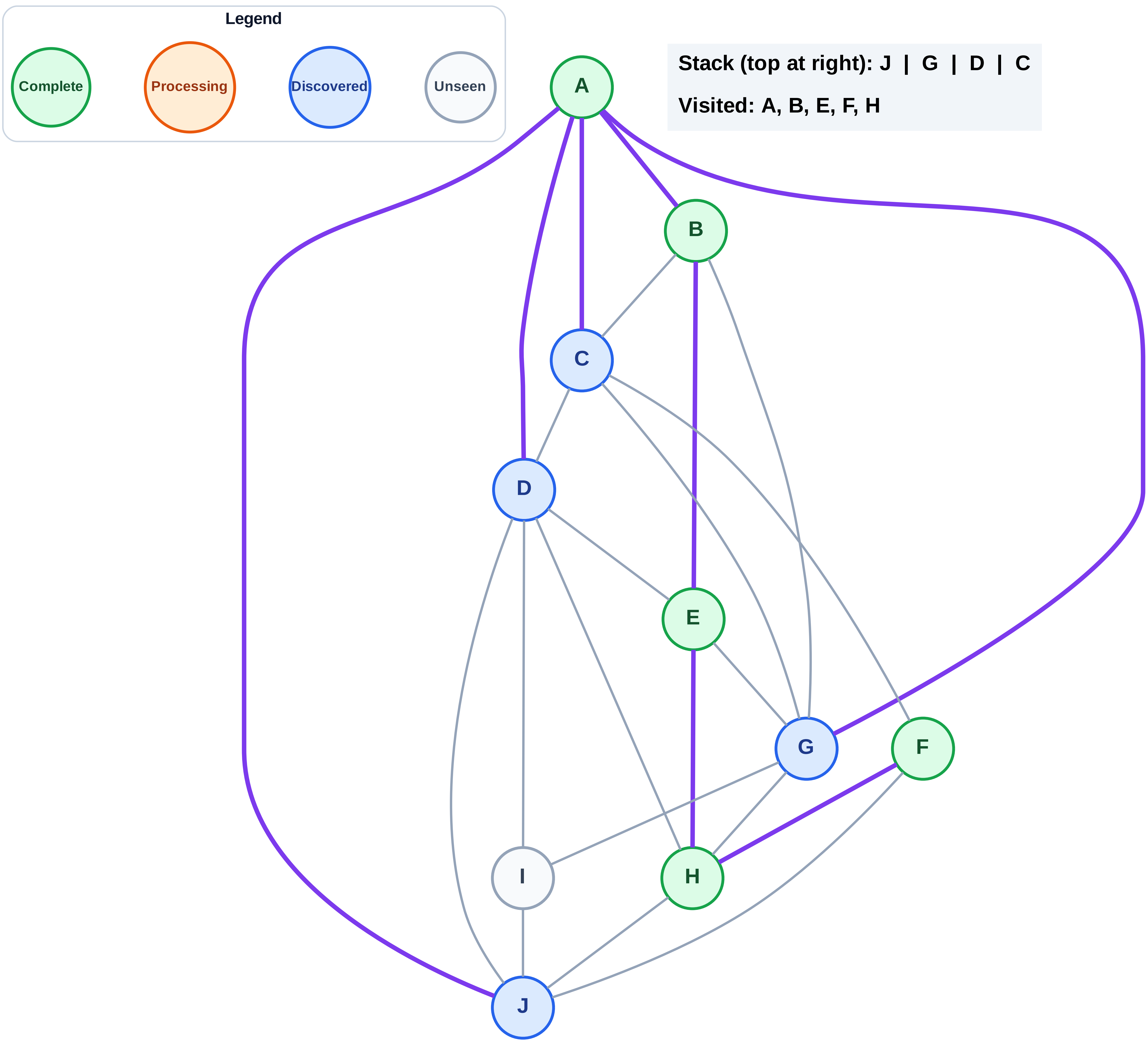
DFS Traversal

Step 10: Process node F



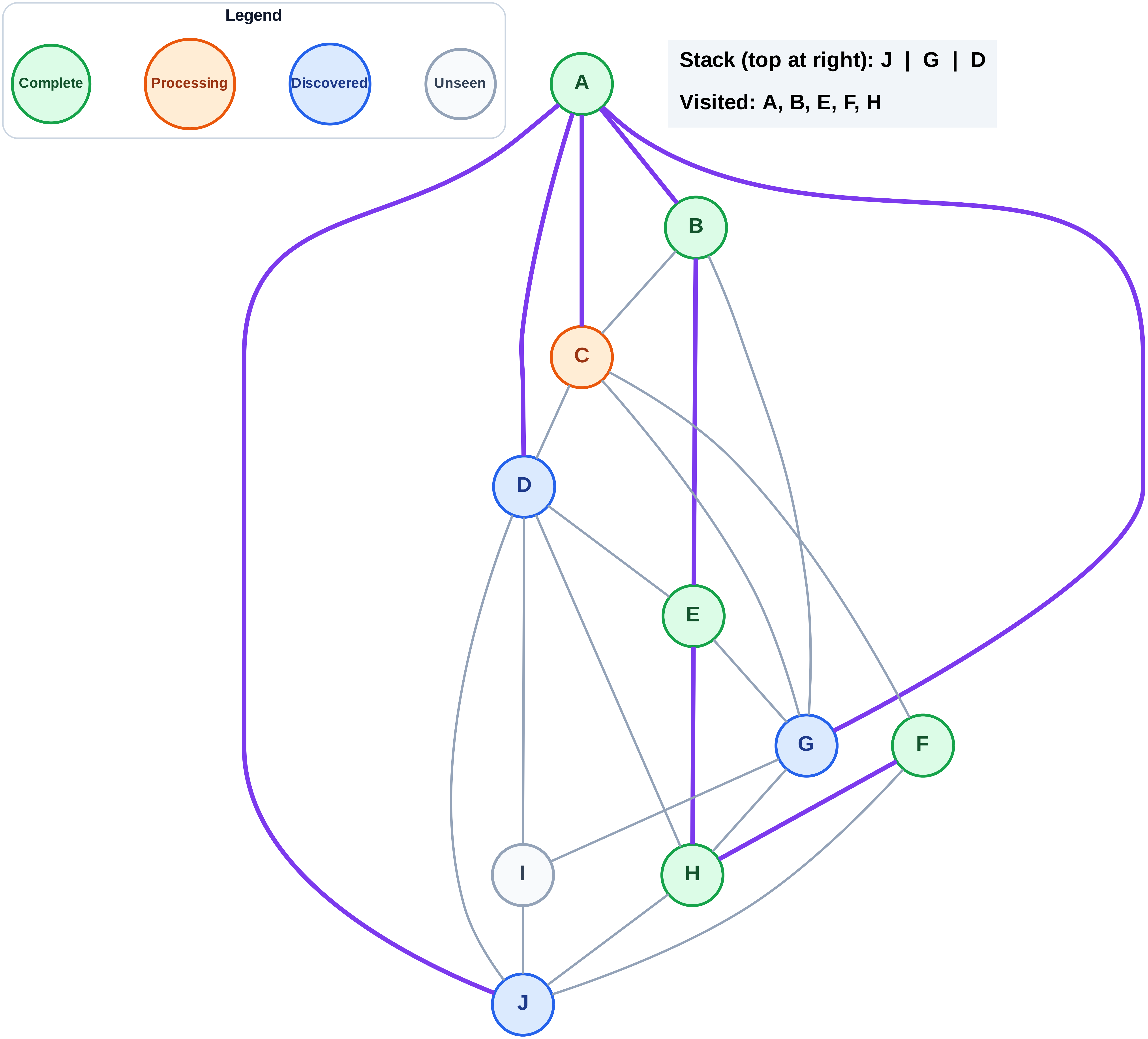
DFS Traversal

Step 11: No new neighbors from F



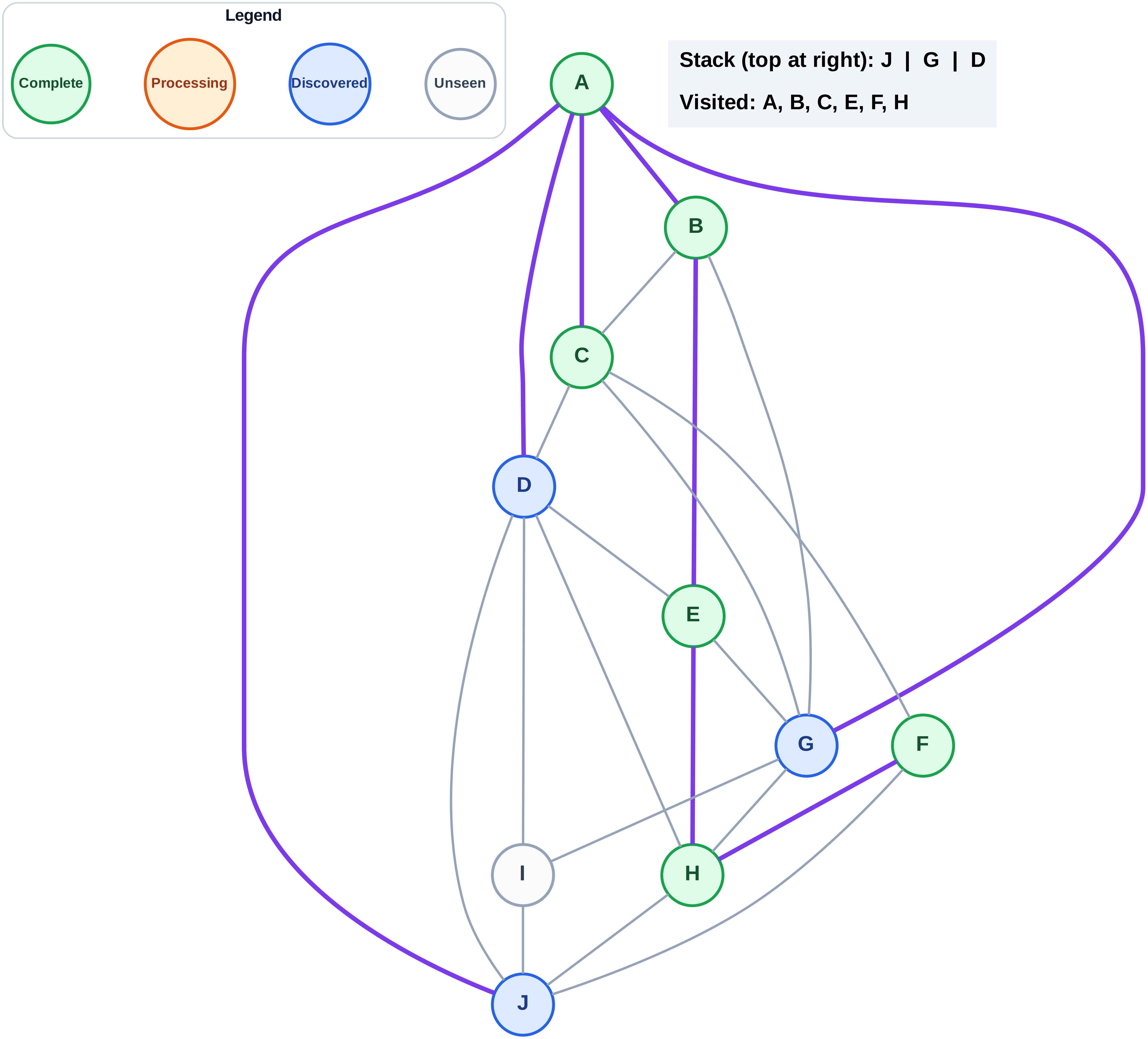
DFS Traversal

Step 12: Process node C



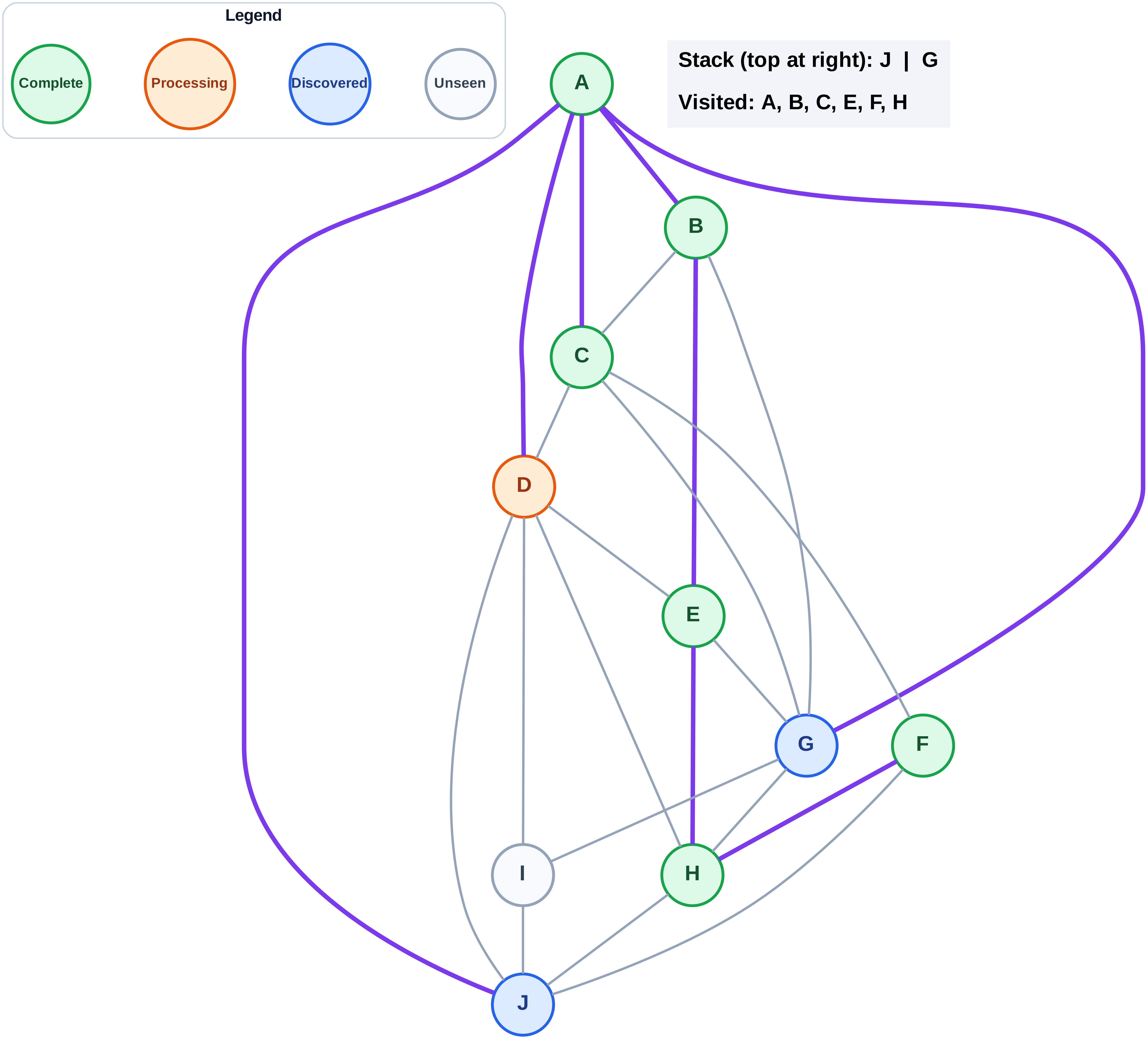
DFS Traversal

Step 13: No new neighbors from C



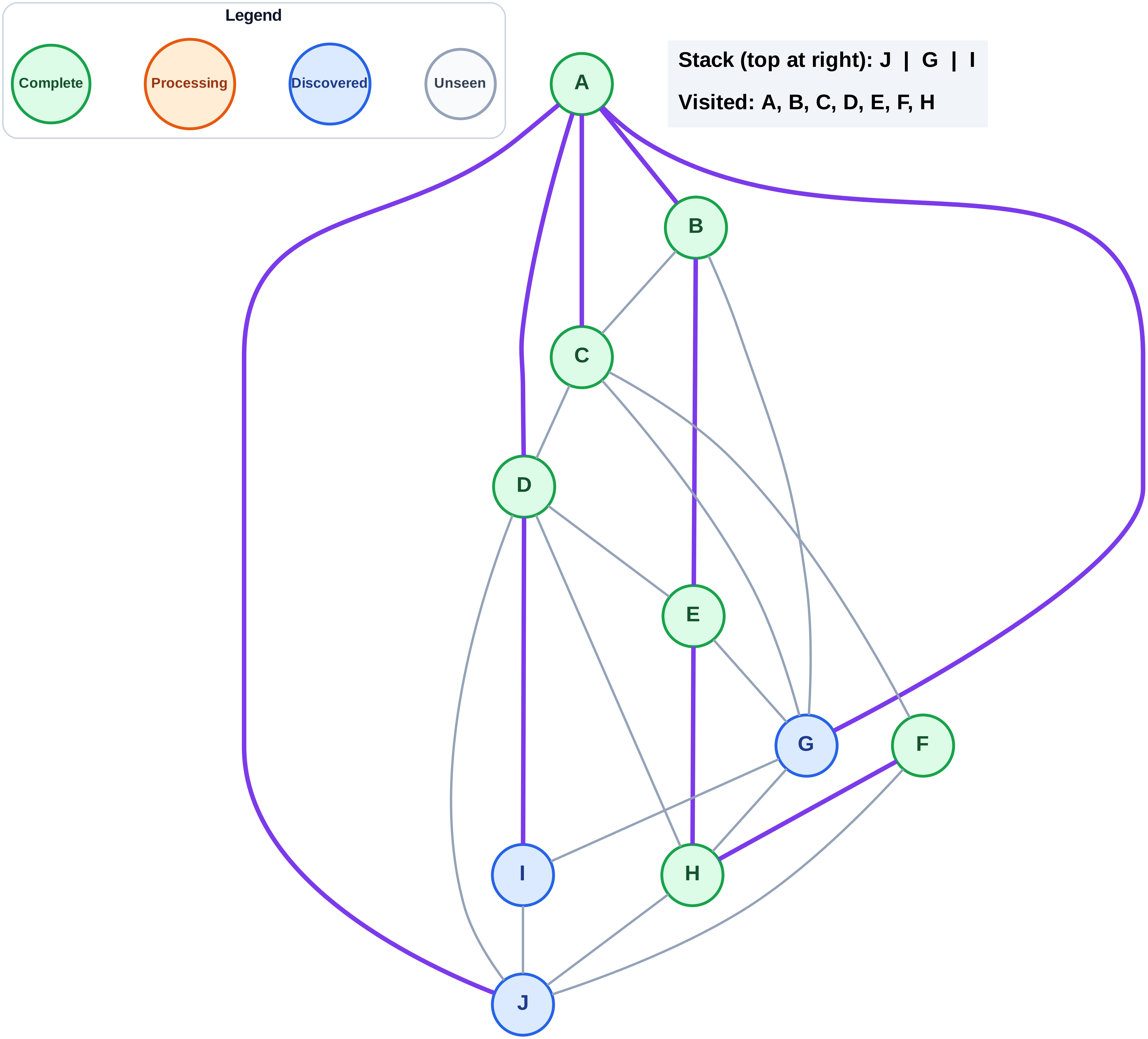
DFS Traversal

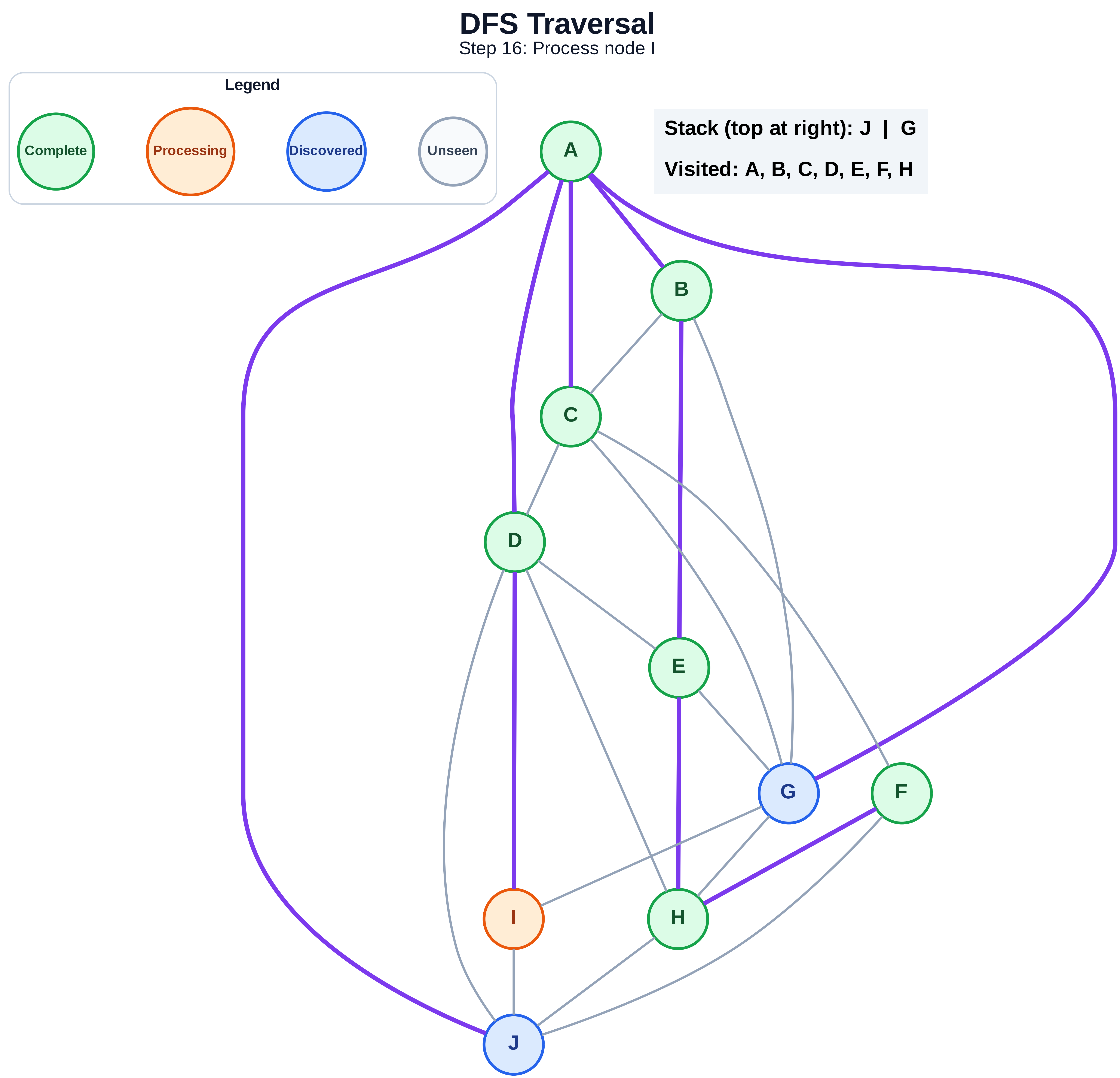
Step 14: Process node D



DFS Traversal

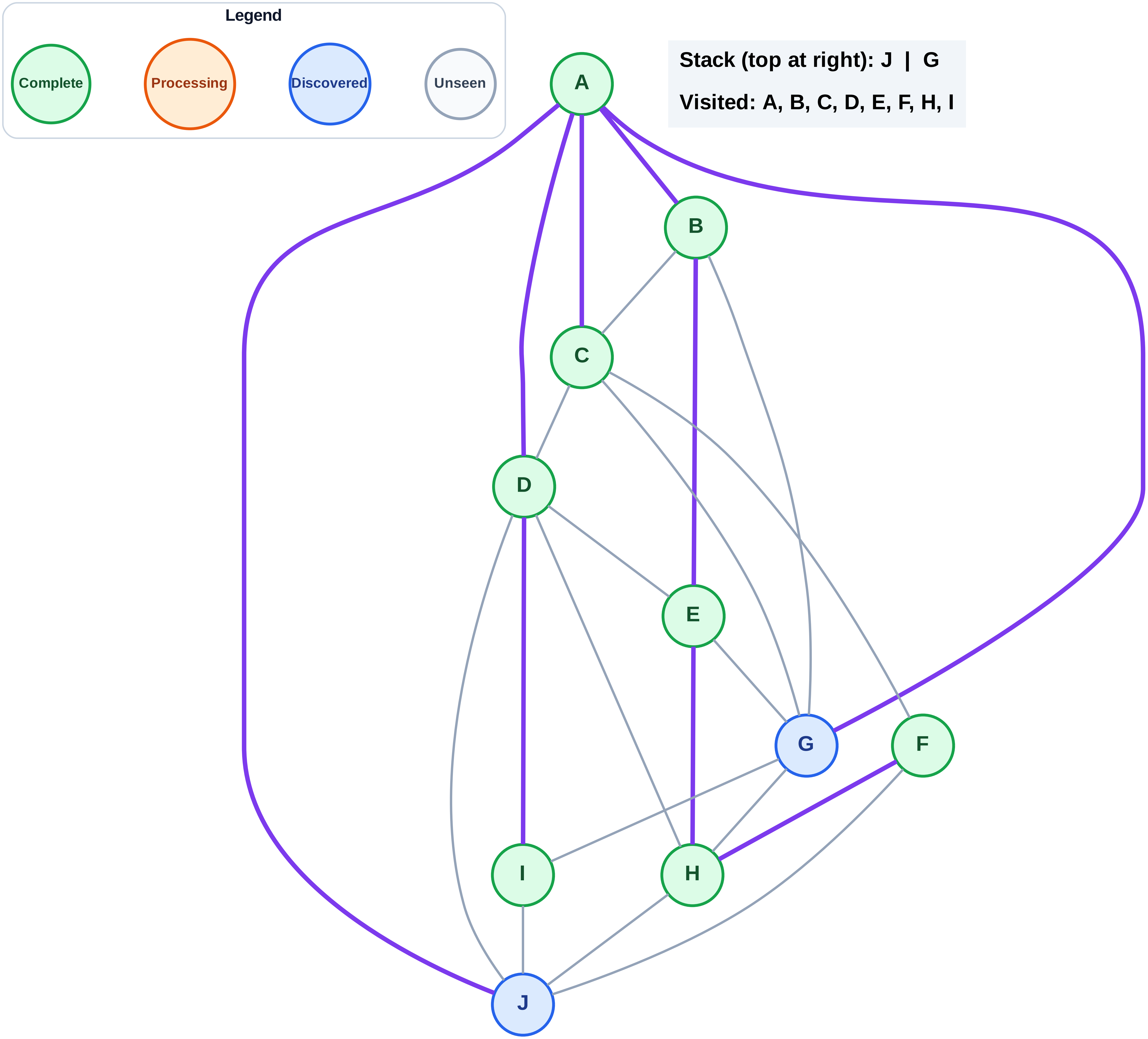
Step 15: Discover I from D




DFS Traversal
Step 16: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G
Visited: A, B, C, D, E, F, H
A
B
C
D
E
F
G
H
I
J

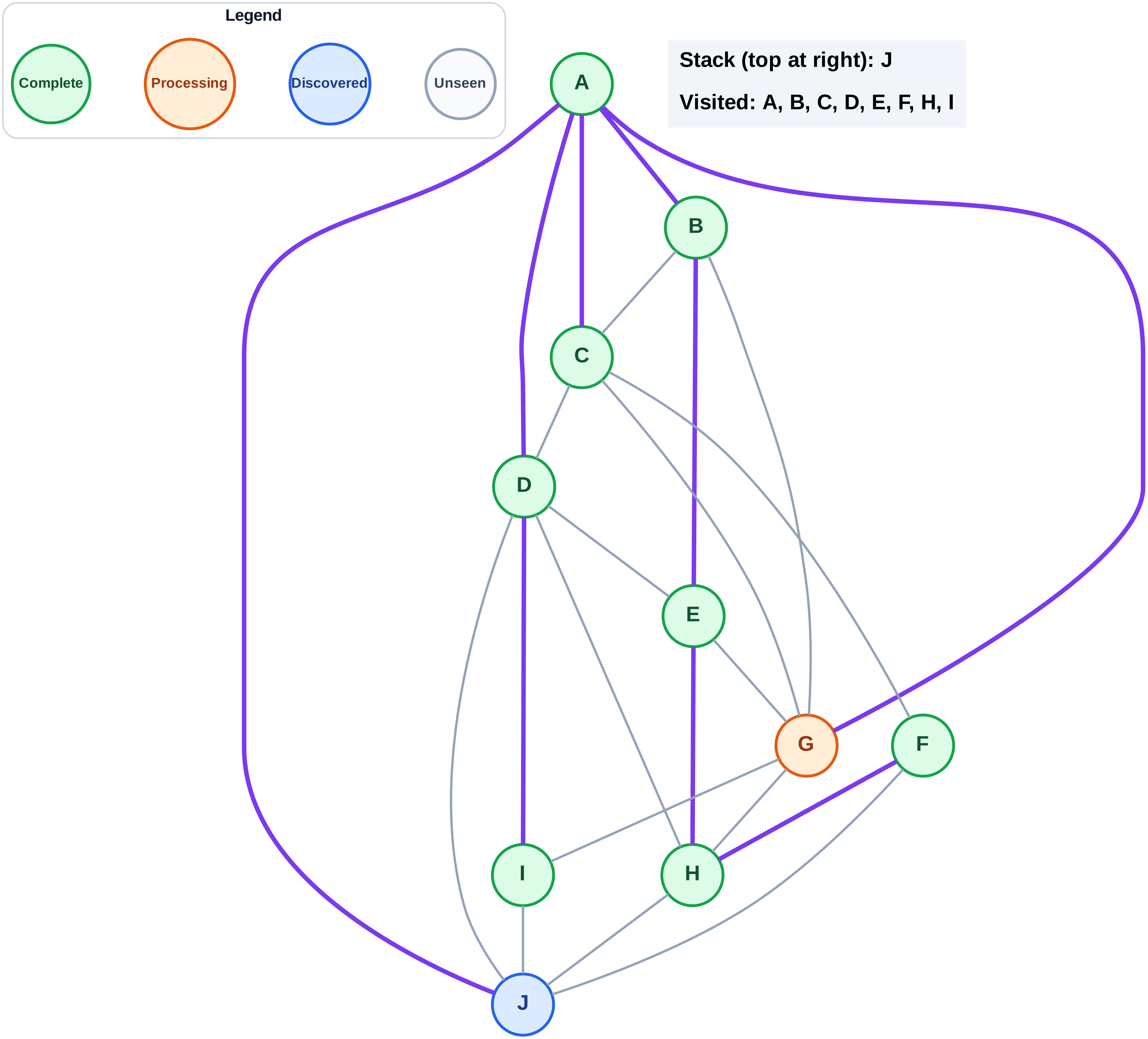
DFS Traversal

Step 17: No new neighbors from I



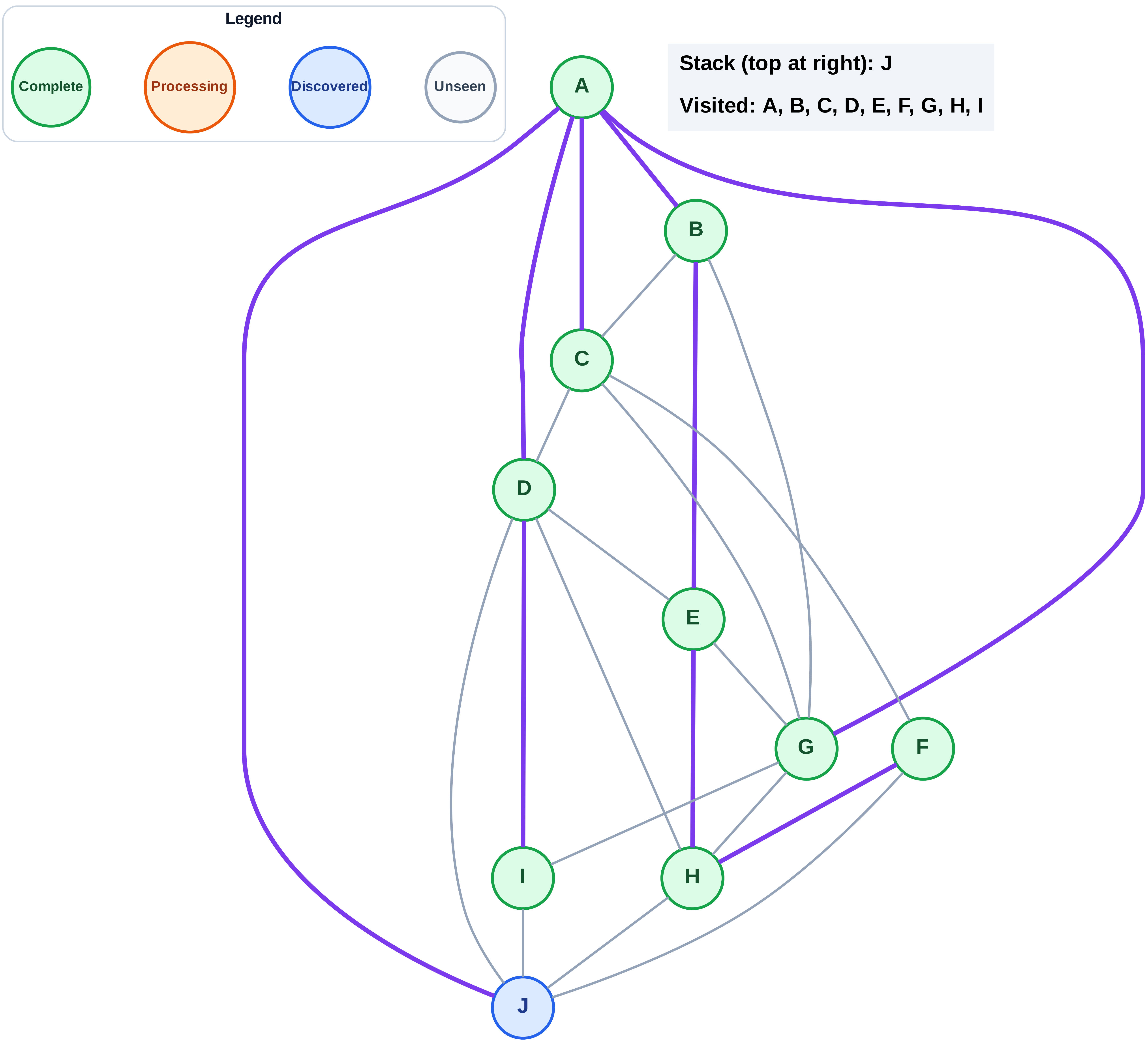
DFS Traversal

Step 18: Process node G



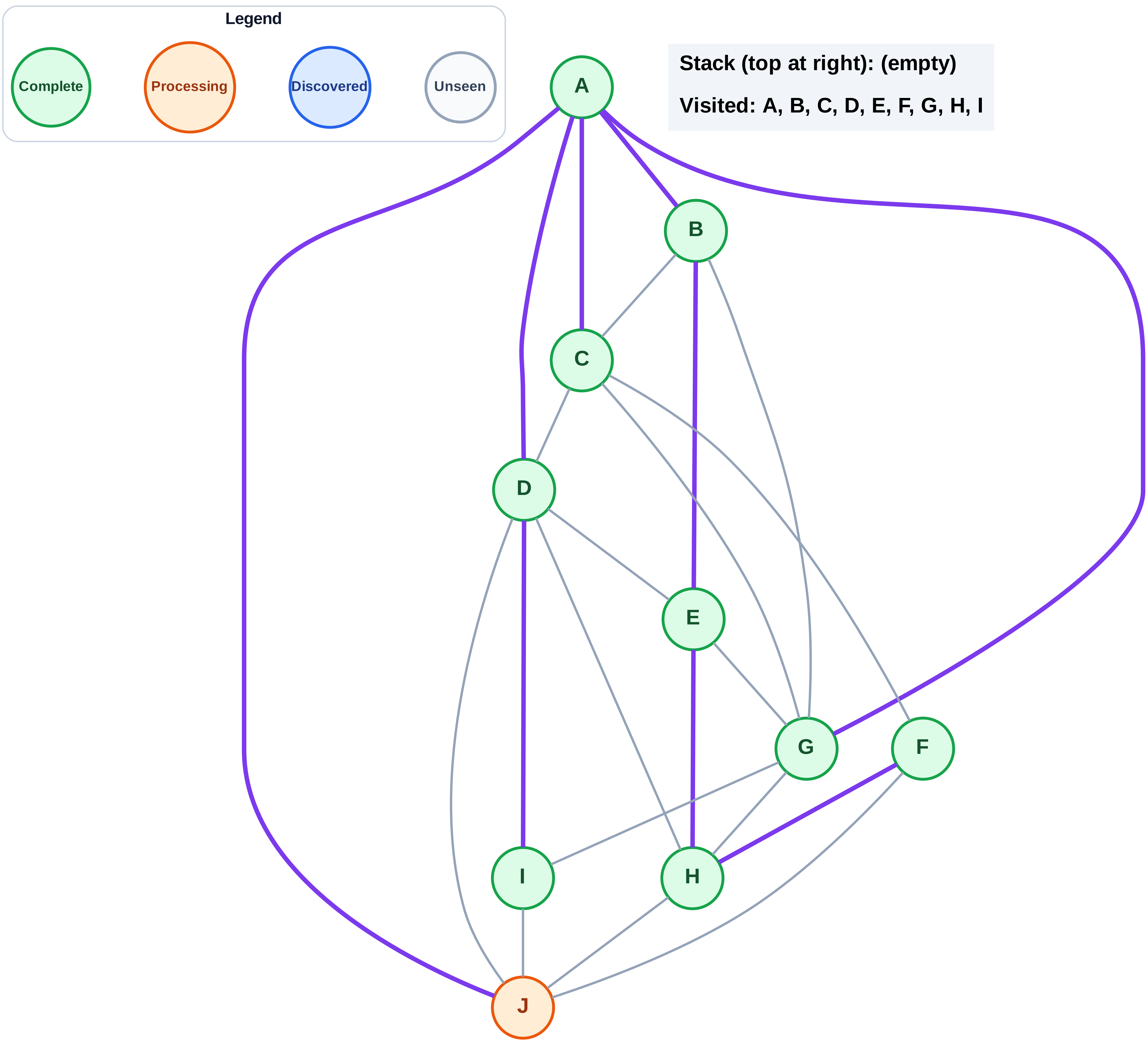
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J